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AIUB, Theory of Computation Notes

THEORY OF COMPUTATION

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INTRODUCTION TO THEORY OF COMPUTATION

“Theory” of “Computation”

⇒ Theory:

- **General & Abstract:** Applies to many systems, independent of technology.
- **Precision:** Used to prove formal mathematical theorems.
- **Positive Results:** Proves what can be computed (e.g., correctness of algorithms).
- **Negative Results:** Proves what cannot be computed (e.g., unsolvable problems with certain costs).

⇒ Computation:

- Any task can be performed by machine that is known as computation.

Automata Theory

→ Automata Overview:

- **Meaning:** From Greek (Αυτόματα) – "self-operating."
- **Study of:** Abstract machines & computational problems (**formal languages**).
- **Role:** Finite representation of infinite sets (**language theory**).
- **Purpose:** Models for computing machines (input → process → output).
- **Use Cases:**
 - **Finite Automaton:** Text processing, compilers design, hardware design.
 - **Context-Free Grammar:** Programming languages, AI.

Computability Theory

→ Computability Theory Overview:

- **Unsolvable Problems:** Some problems cannot be solved by computers.
- **Example:** Determining if a mathematical statement is true or false (undecidable).
- **Theoretical Models:** Ideas that led to actual computer development.
- **Problem Classification:** Divides problems into solvable and unsolvable categories.

Complexity Theory

→ Computability & Complexity Theory

- **Computability Theory:** Determines if a problem is **Solvable / Not Solvable**.
- **Complexity Theory:** If solvable, classifies it as **Easy / Hard**.
- **Problem Classification:**
 - Modify the problem to make it easier.
 - Settle for approximate solutions.
 - Some problems are hard only in worst-case scenarios.
- **Cryptography:** Requires problems that are **hard to solve** without a key.

Summary

Theory	Concept	Examples / Applications
Computability Theory	Determines if a problem is Solvable / Not Solvable	If solvable → moves to Complexity Theory
Complexity Theory	Classifies solvable problems as Easy / Hard	- Easy: Sorting (e.g., arranging millions of numbers) - Hard: Scheduling (e.g., conflict-free class routine)
Automata Theory	Deals with Mathematical Models	Classification: - Finite Automaton: Used in text processing, compiler design, hardware design - Context-Free Grammar: Used in programming languages, AI

DETERMINISTIC FINITE AUTOMATON (DFA)

Automata Theory

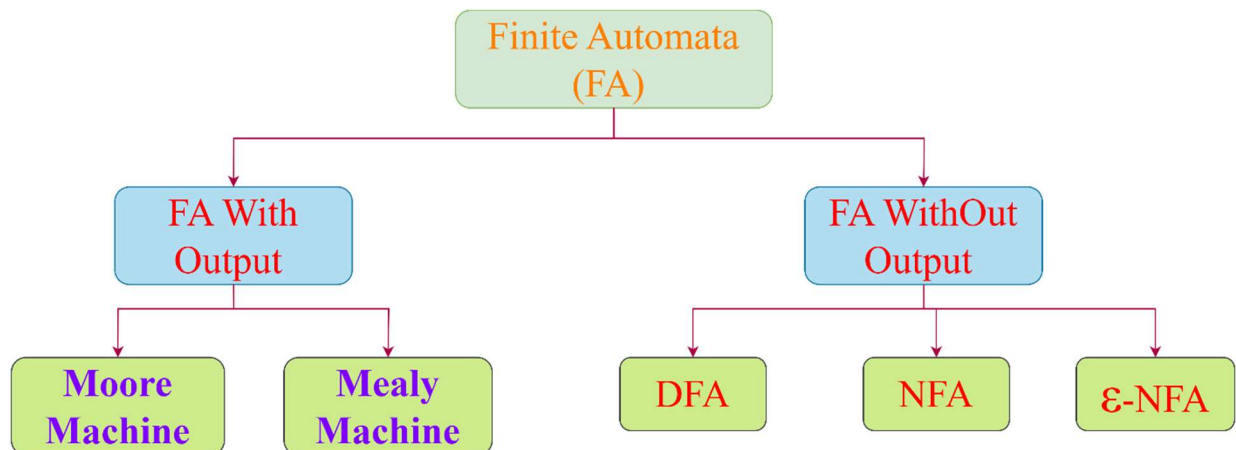
Concept	Description	Examples / Applications
Origin	Comes from the Greek word "Αυτόματα" meaning "self-operating"	—
Definition	Study of abstract mathematical machines and computational problems	—
Purpose	Theoretical models for computing machines (Input → Process → Output)	Used for proofs about computability
Formal Languages	Automaton represents infinite language sets	Formal languages specify problems for computation
Types of Automata	- Finite Automaton - Context-Free Grammar	- Finite Automaton: Used in text processing, compiler design, hardware design - Context-Free Grammar: Used in programming languages, AI

Terminology/Prerequisites

Concept	Definition	Examples
Symbol	A single character or element from an alphabet.	$a, b, c, \dots z; 0, 1, 2, \dots 9; A, 3, f, 5, \dots$
Alphabet (Σ)	A non-empty finite set of symbols, denoted by Σ .	$\Sigma = \{a, b\}, \Sigma = \{0, 1\}, \Sigma = \{0, 1, \dots 9\}$
String (w)	A finite sequence of symbols from an alphabet.	$\Sigma = \{a, b\}, w = ab, ba$

Concept	Definition	Examples
Language (L)	A collection of all possible strings over an alphabet.	Finite Example: $\Sigma = \{a, b\}$, $L_1 = \{\text{Set of all strings of length 2}\} = \{aa, ab, bb, ba\}$ $L_2 = \{\text{Set of all strings of length 3}\} = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$ Infinite Example: $\Sigma = \{a, b\}$, $L_2 = \{\text{Set of all strings starting with 'a'}\} = \{a, aa, ab, abb, aba, \dots\}$
Power of Alphabet (Σ^n)	Represents all possible strings of a given length over an alphabet.	$\Sigma^0 = \{\epsilon\}$ (empty string) $\Sigma^1 = \{a, b\}$ (strings of length 1) $\Sigma^2 = \{aa, ab, ba, bb\}$ (strings of length 2) $\Sigma^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$ (strings of length 3) $\Sigma = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$ (all possible strings of any length)

Finite State Machine



Finite Automata

➔ Finite Automaton Overview:

- **Simple Model:** Used for devices with limited memory.
- **Common Applications:** Dishwashers, electronic thermostats, digital watches, calculators.
- **Key Focus:** Limited states and transitions based on input.
- **Goal:** Design and understand the behavior of devices using finite memory.

➔ Finite Representation of an Infinite Language

Finite Example:

- **Alphabet:** $\Sigma = \{a, b\}$
- **Language ($L(M)$):** Set of all strings of length 2
 - $L(M) = \{aa, ab, bb, ba\}$
 - **Accepted String:** "ab" (Machine will accept)
 - **Rejected String:** "abc" (Machine will not accept)

Infinite Example:

- **Alphabet:** $\Sigma = \{a, b\}$
- **Language ($L(M)$):** Set of all strings starting with 'a'
 - $L(M) = \{a, aa, ab, abb, aba, \dots\}$
 - **Machine Behavior:** Will search indefinitely for strings like "baaaa."

➔ **Problem:** An infinite language requires infinite time for the machine to search through every possible string. This is impractical.

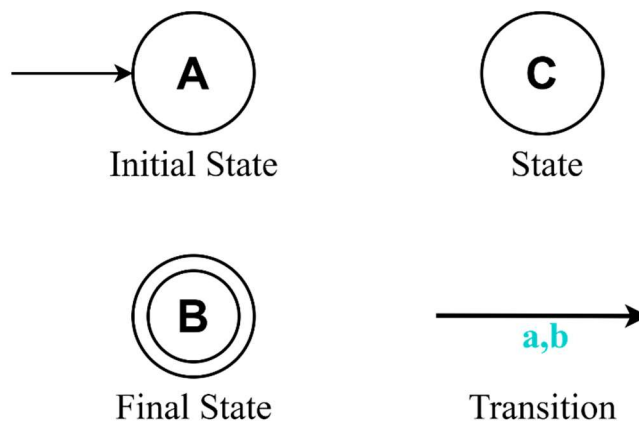
➔ Solution: Finite Representation (FR)

- **Goal:** Use a finite model (e.g., regular expression, finite automaton) to represent infinite languages.
 - **Example FR:** $a(a|b)^*$ for strings starting with 'a'.
 - **Approach:** Use compact representations (like regular expressions) to avoid infinite search.

Finite Representation (FR) / Finite Automaton (FA)

- **Definition:** If a language can be recognized with a finite number of states, it's called Finite Representation (FR) or Finite Automaton (FA).
- **Key Point:** FA uses a finite number of states to represent and accept or reject strings from a language.

➔ Finite Representation:

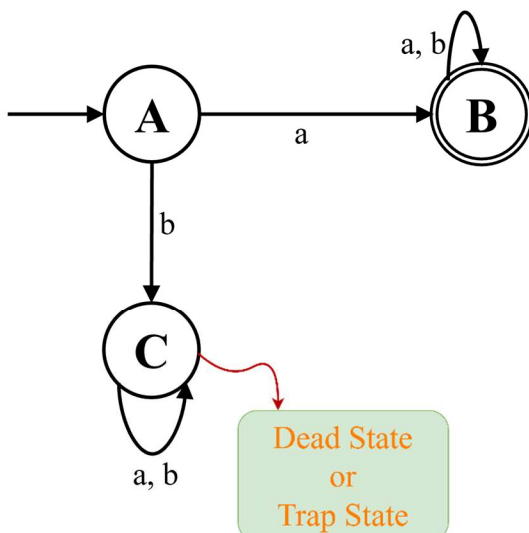


🔗 Example 1:

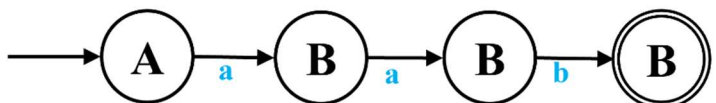
➔ $\Sigma = \{a, b\}$

➔ $L1 = \text{Set of all strings starting with 'a'}$

➔ $L1 = \{a, aa, ab, abb, aba, \dots\}$



ACCEPT:

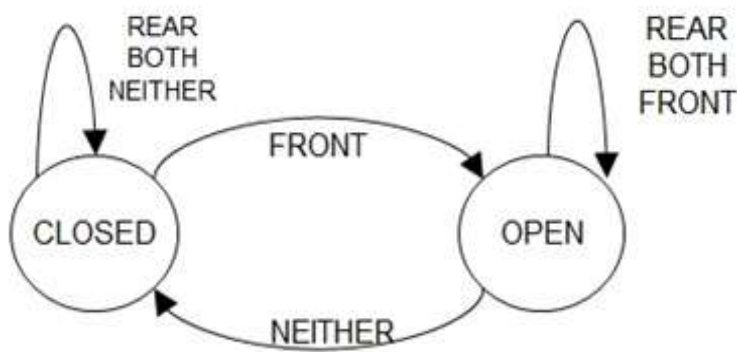


REJECT:



Parts of a Finite Automaton (FA)

✂ Example:



		Input Signal			
State		Front	Rare	Both	Neither
	Closed	Open	Closed	Closed	Closed
	Open	Open	Open	Open	Closed

Fig: State Transition for Automatic Door Controller

1. **Input (Language):** Defined set of inputs.
 - **Example:** FRONT, REAR, BOTH, NEITHER.
2. **Set of States:** Different conditions the system can be in.
 - **Example:** OPEN, CLOSED (for an automatic door).
3. **Initial (Start) State:** The starting condition.
 - **Example:** CLOSED (default state of the door).
4. **Transition Rules:** Define state changes based on input.
 - **Example:** If in CLOSED state and input is FRONT, move to OPEN state.
5. **Final/Accept States:** The goal state(s).
 - **Example:** CLOSED when the last input is NEITHER.

Deterministic Finite Automaton (DFA)

1. Deterministic Computation:

- Each step follows uniquely from the previous one.
- Given a state and input, the next state is determined.

2. Transition Rules:

- Each state has exactly **one** transition per input.
- If there are **n** inputs, each state has **n** transitions.

3. Key Properties:

- **One Start State** $\rightarrow$ Begins the transition.
- **One or More Final States** $\rightarrow$ Marks the completion.

Formal Definition - DFA

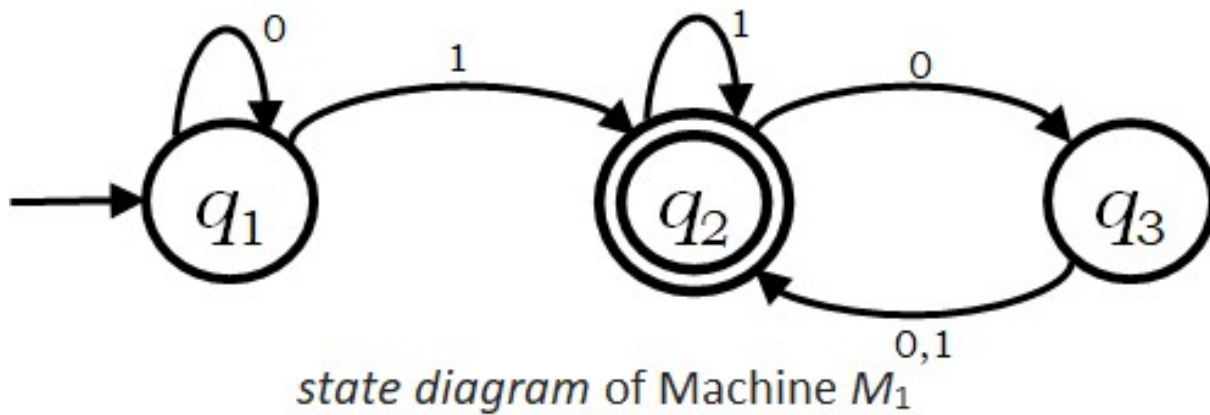
$\rightarrow$ A DFA is a 5-tuple: $(Q, \Sigma, \delta, q_0, F)$

1. $Q \rightarrow$ Finite set of states.
2. $\Sigma \rightarrow$ Finite input alphabet.
3. $\delta: Q \times \Sigma \rightarrow Q \rightarrow$ Transition function (maps state & input to next state).
4. $q_0 \in Q \rightarrow$ Start state.
5. $F \subseteq Q \rightarrow$ Set of accept (final) states.

Language Recognition:

- If **A** is the set of strings accepted by DFA machine **M**, then:
 - **$L(M) = A \rightarrow$ DFA recognizes A.**
 - **M accepts A** if it reaches a final state after reading **input**.

☞ Example 1:



→ Here,



$$M_1 = (Q, \Sigma, d, q_0, F),$$

➤ Where:

$$Q = \{q_1, q_2, q_3\},$$

$$\Sigma = \{0, 1\},$$

$$q_0 = \{q_1\},$$

$$F = \{q_2\},$$

δ is describe as:

$$\delta(q_1, 0) = q_1, \quad \delta(q_1, 1) = q_2,$$

$$\delta(q_2, 0) = q_3, \quad \delta(q_2, 1) = q_2,$$

$$\delta(q_3, 0) = q_2, \quad \delta(q_3, 1) = q_2.$$

Transition Function

or

δ	0	1
q_1	q_1	q_2
q_2	q_3	q_2
q_3	q_2	q_2

Transition Table

Mathematical Formalization of DFA Computation

⇒ Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA.

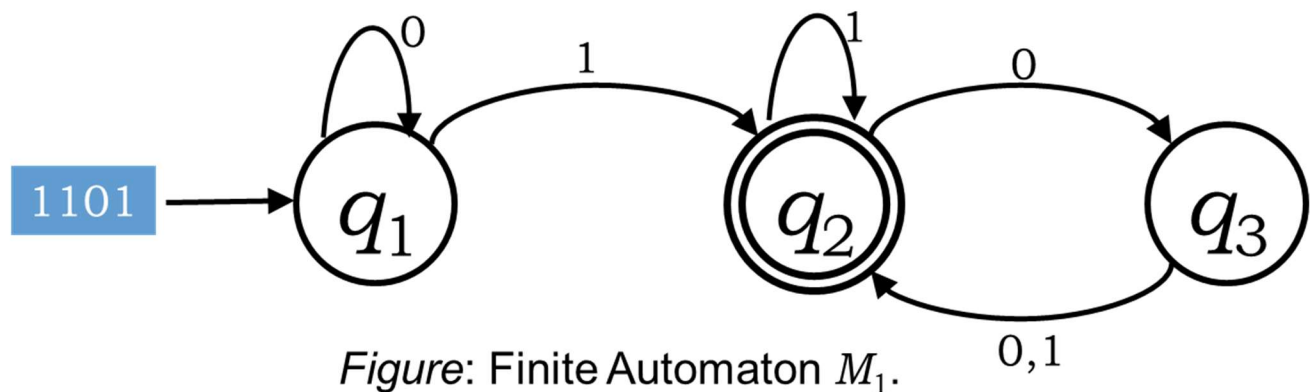
For a string $w = w_1 w_2 \dots w_n \in \Sigma^*$ (where each $w_i \in \Sigma$), M accepts w if a sequence of states $r_0, r_1, \dots, r_n$ exists in Q , satisfying:

1. **Start State:** $r_0 = q_0$
2. **State Transition:** $\delta(r_i, w_{i+1}) = r_{i+1}$ for $i = 0, 1, \dots, n-1$
3. **Final State:** $r_n \in F$

Language Recognition:

- M recognizes language L if $L = \{w \mid M \text{ accepts } w\}$ (i.e., all strings leading to a final state).

Example 2:



→ We know:

➤ $M = (Q, \Sigma, \delta, q_0, F)$,

➤ Where:

- ✓ $Q = \{q_1, q_2, q_3\}$,
- ✓ $\Sigma = \{0, 1\}$,
- ✓ $q_0 = q_1$,
- ✓ $F = \{q_2\}$,

$\delta =$

$\delta(q_1, 0) = q_1,$	$\delta(q_1, 1) = q_2,$
$\delta(q_2, 0) = q_3,$	$\delta(q_2, 1) = q_2,$
$\delta(q_3, 0) = q_2,$	$\delta(q_3, 1) = q_2.$

- Input string $w = w_1w_2w_3w_4 = 1101$ to M_1 gives a sequence of states r_0, r_1, r_2, r_3, r_4 in the following computation (here $n = 4$) –
- Start in state $r_0 = q_1$; $\rightarrow r_0 = q_0$,
- $\delta(r_0, w_1) = \delta(q_1, 1) = q_2 = r_1$; $\rightarrow \delta(r_i, w_{i+1}) = r_{i+1}$, for $i = 0, 1, 2, \dots, n-1$,
- $\delta(r_1, w_2) = \delta(q_2, 1) = q_2 = r_2$; $\rightarrow \delta(r_i, w_{i+1}) = r_{i+1}$, for $i = 0, 1, 2, \dots, n-1$,
- $\delta(r_2, w_3) = \delta(q_2, 0) = q_3 = r_3$; $\rightarrow \delta(r_i, w_{i+1}) = r_{i+1}$, for $i = 0, 1, 2, \dots, n-1$,
- $\delta(r_3, w_4) = \delta(q_3, 1) = q_2 = r_4$; $\rightarrow r_n \in F$.

Step	Current State (r_i)	Input Symbol (w_{i+1})	Next State (r_{i+1})	Transition Used (δ)
0	q1 (Start)	1	q2	$\delta(q_1, 1) = q_2$
1	q2	1	q2	$\delta(q_2, 1) = q_2$
2	q2	0	q3	$\delta(q_2, 0) = q_3$
3	q3	1	q2	$\delta(q_3, 1) = q_2$
4	q2 (Accept)	-	-	w is Accepted

Conclusion:

- The input $w = 1101$ leads to **q2**, which is an **accept state**.
- M_1 accepts w , meaning $w \in L(M_1)$.
- Language $L = \{w \mid M_1 \text{ accepts } w\}$.

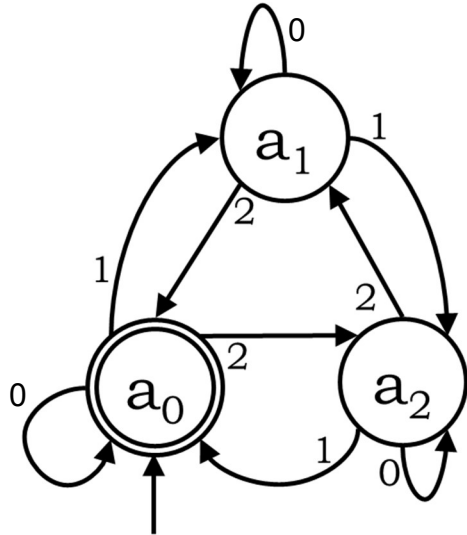
DFA design issues

- Identify the **pattern/type** of input strings in the language.
- Match input pattern **step-by-step** with states & transitions.
- Ensure **necessary conditions** before completing the state diagram.
- **Rules to follow:**
 - Each state must have **one transition per input symbol**.
 - **Self-loops** allow zero or more occurrences of an input symbol.
 - There can be **one or more final states**.
- Create **test input strings** (both **ACCEPTED & REJECTED**).
- Test strings **while drawing/updating** the state diagram.

Designing DFA

✂ Example 1:

➔ Design a DFA that accepts strings where the sum of all symbols (0,1,2) is a multiple of 3.



Input example: 01120101

Input symbol: Accepted

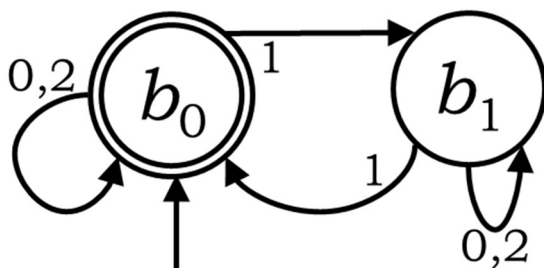
➔ DFA for Sum of Symbols $\equiv 0,1,2 \pmod 3$:

- **Alphabet:** $\Sigma = \{0,1,2\}$
- **Language:** Strings where the sum of symbols is a multiple of 3.
- **States:** $Q = \{a_0, a_1, a_2\}$, where a_i represents $S \bmod 3 = i$.
- **Start State:** $q_0 = a_0$
- **Final State:** $F = \{a_0\}$
- **Transition Table (δ_1):**

δ	0	1	2
a_0	a_0	a_1	a_2
a_1	a_1	a_2	a_0
a_2	a_2	a_0	a_1

✂ Example 2:

➔ Design a DFA that accepts strings where the sum of all symbols (0,1,2) is an even number.



Input example: 01120101

Input symbol: Accepted

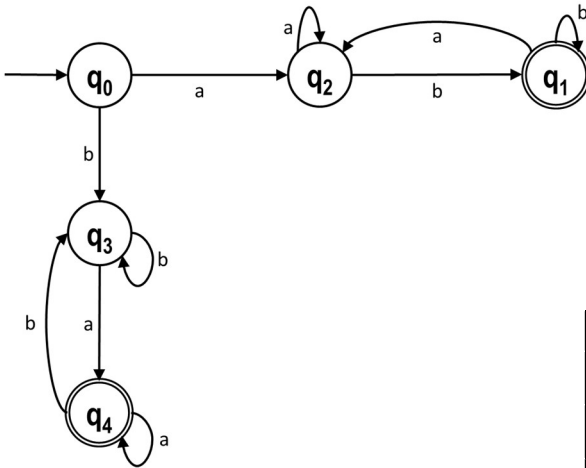
DFA for Sum of Symbols $\equiv 0,1,2 \pmod 2$ (Even Sum):

- **Alphabet:** $\Sigma = \{0,1,2\}$
- **Language:** Strings where the sum of symbols is an even number.
- **States:** $Q = \{b_0, b_1\}$, where b_i represents $S \bmod 2 = i$.
- **Start State:** $q_0 = b_0$
- **Final State:** $F = \{b_0\}$
- **Transition Table (δ_2):**

δ	0	1	2
b_0	b_0	b_1	b_0
b_1	b_1	b_0	b_1

Example 3:

→ Identify the Components of the DFA



Components of the DFA:

- **Alphabet:** $\Sigma = \{a, b\}$
- **States:** $Q = \{q_0, q_1, q_2, q_3, q_4\}$,
- **Start State:** $q_0 = q_0$
- **Final State:** $F = \{q_1, q_4\}$
- **Transition Table (δ):**

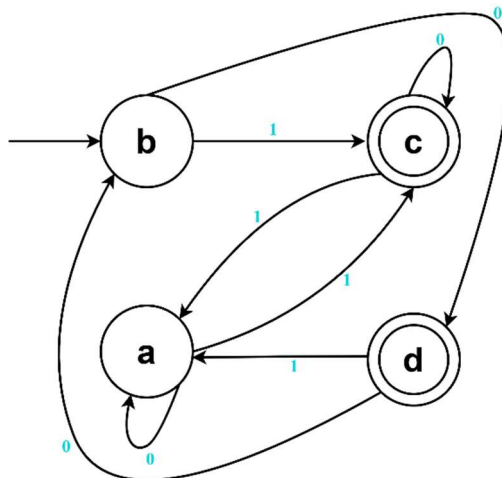
δ	a	b
q_0	q_2	q_3
q_1	q_2	q_1
q_2	q_2	q_1
q_3	q_4	q_3
q_4	q_4	q_3

Example 4:

→ Design a DFA that $Q = \{a, b, c, d\}$, $\Sigma = \{0, 1\}$, $q_0 = b$, $F = \{c, d\}$,

δ	0	1
a	a	c
b	d	c
c	c	a
d	b	a

→ **Solution:**



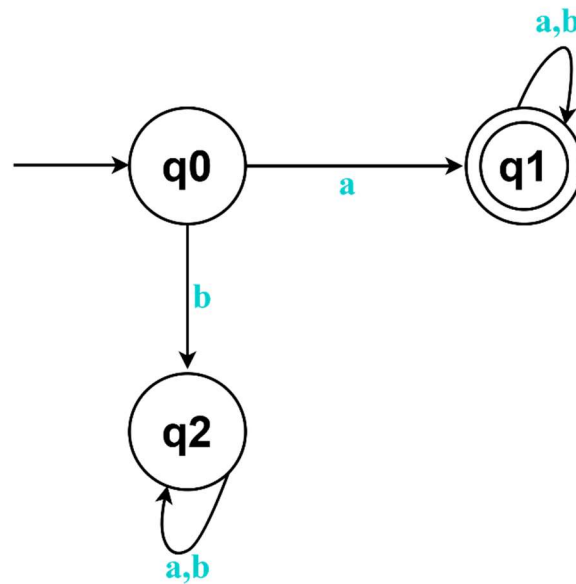
✂ Example 5:

→ Given the state diagram of the machine ($\{q_0, q_1, q_2\}, \{a, b\}, \delta, q_0, \{q_1\}$) Where δ is given the following table

δ	a	b
q_0	q_1	q_2
q_1	q_1	q_1
q_2	q_2	q_2

→ Design The DFA

→ **Solution:**

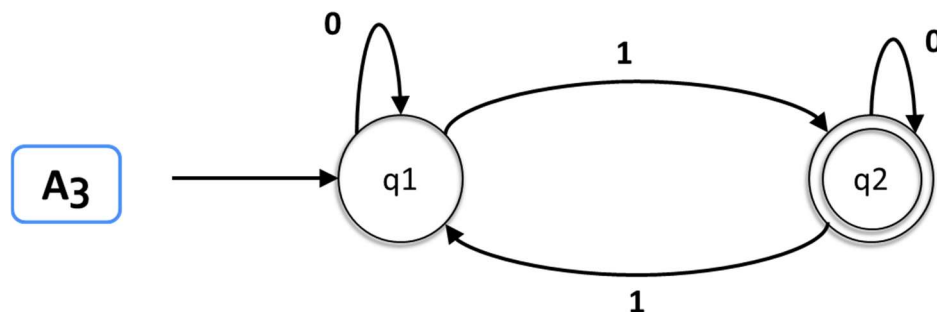


✂ Example 6:

→ $A_3 = \{w : w \text{ is a binary string containing an odd number of 1s}\}. \Sigma = \{0,1\}$

→ So, $w_1 = 1, w_2 = 10, w_3 = 101, w_4 = 10101, w_5 = 10101\dots$

Now, $L = \{1, 100, 10101, 11010, 11100\dots\}$



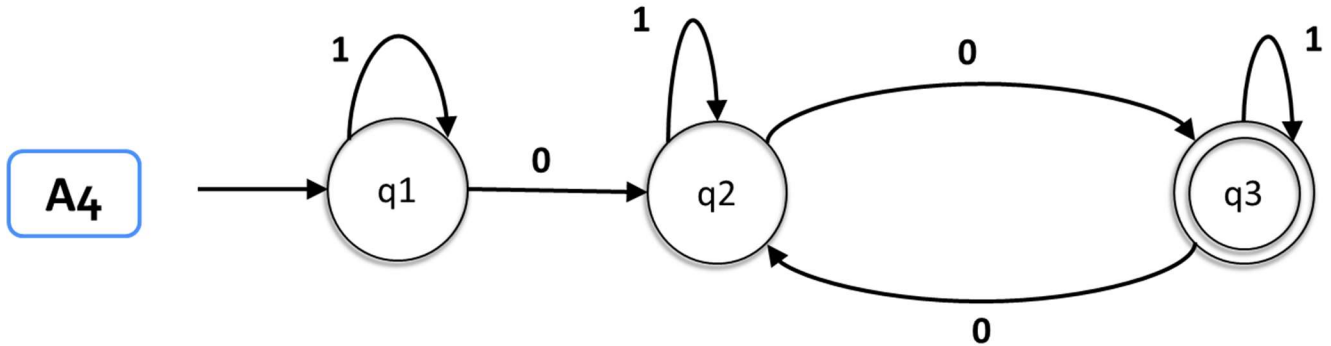
Accepted

✂ Example 7:

→ $A_4 = \{w : w \text{ is a binary string containing an even number of 0s}\}. \Sigma = \{0,1\}$

→ So, $w_1 = 00, w_2 = 001, w_3 = 0101, w_4 = 01010, w_5 = 0101010...$

Now, $L = \{00, 100, 1010, 11010, 11100...\}$

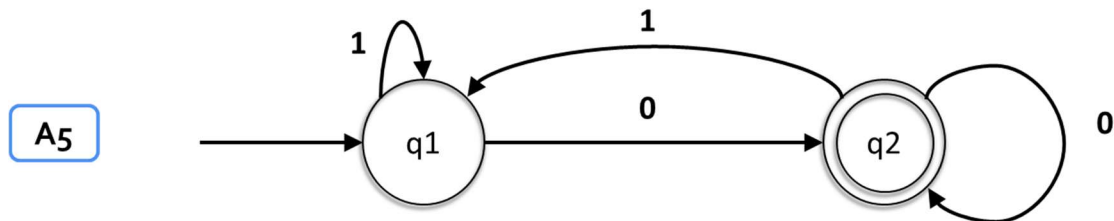


✂ Example 8:

→ $A_5 = \{w \mid w \text{ ends with a 0}\}. \Sigma = \{0,1\}$

→ So, $w_1 = 010, w_2 = 000, w_3 = 10, w_4 = 010101, w_5 = 0101010...$

Now, $L = \{0, 10, 100, 11010, 11100...\}$

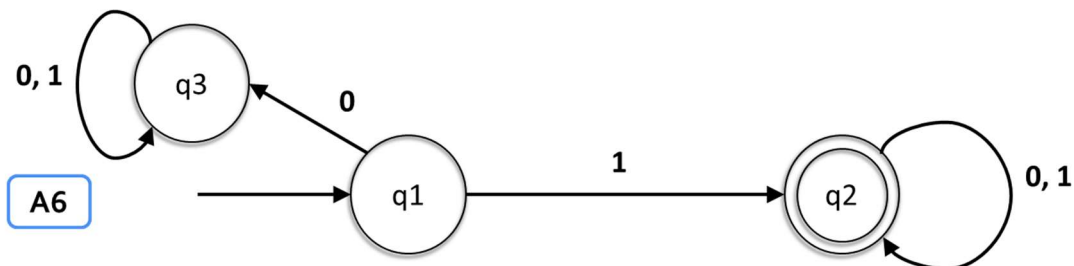


✂ Example 9:

→ $A_6 = \{w \mid w \text{ begins with a 1}\}. \Sigma = \{0,1\}$

→ So, $w_1 = 100, w_2 = 001, w_3 = 10, w_4 = 01010, w_5 = 101010...$

Now, $L = \{1, 11, 101, 1001, 11100...\}$

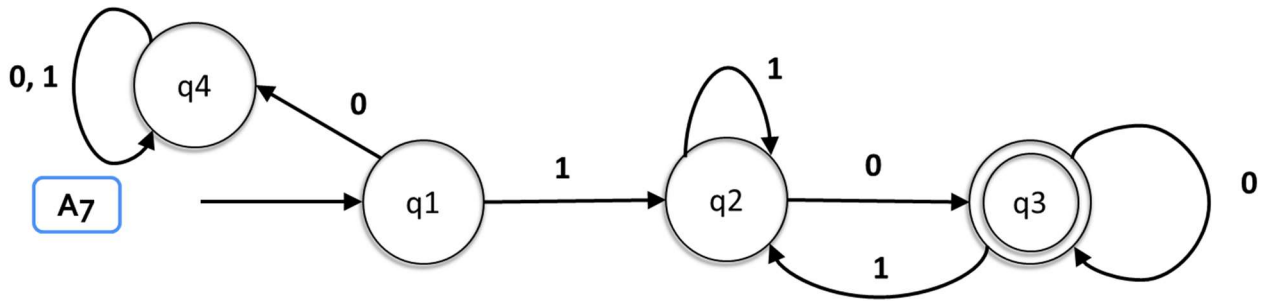


✂ Example 10:

→ $A7 = \{w \mid w \text{ begins with a } 1 \text{ and ends with a } 0\}$. $\Sigma = \{0,1\}$

→ So, $w_1 = 100$, $w_2 = 001$, $w_3 = 10$, $w_4 = 01010$, $w_5 = 101010...$

Now, $L = \{1, 100, 1010, 11010, 11100...\}$

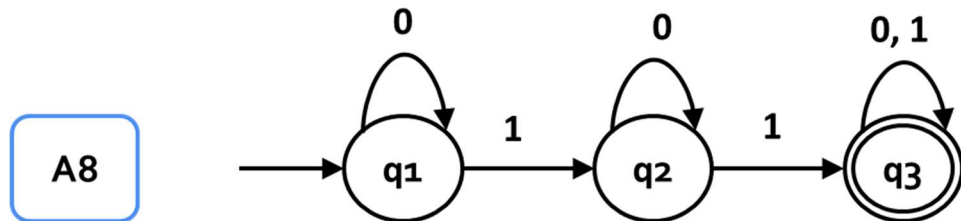


✂ Example 11:

→ $A8 = \{w \mid w \text{ has at least two } 1\text{s}\}$. $\Sigma = \{0,1\}$

→ So, $w_1 = 1010$, $w_2 = 001$, $w_3 = 101$, $w_4 = 01010$, $w_5 = 101010...$

Now, $L = \{11, 1100, 1010, 11010, 11100...\}$

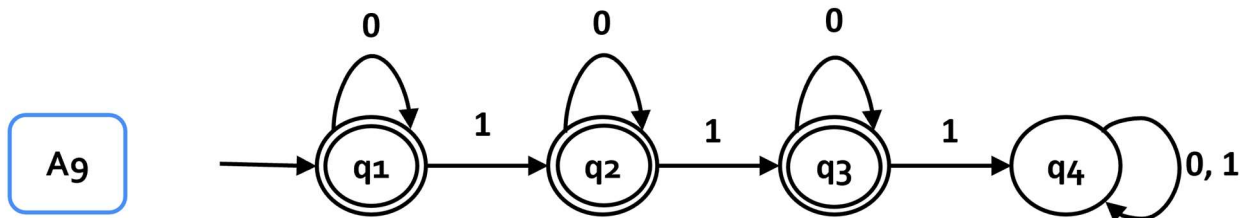


✂ Example 12:

→ $A9 = \{w \mid w \text{ has at most two } 1\text{s}\}$. $\Sigma = \{0,1\}$

→ So, $w_1 = 1010$, $w_2 = 001$, $w_3 = 000$, $w_4 = 010110$, $w_5 = 10100...$

Now, $L = \{11, 1100, 1010, 11010, 11100...\}$

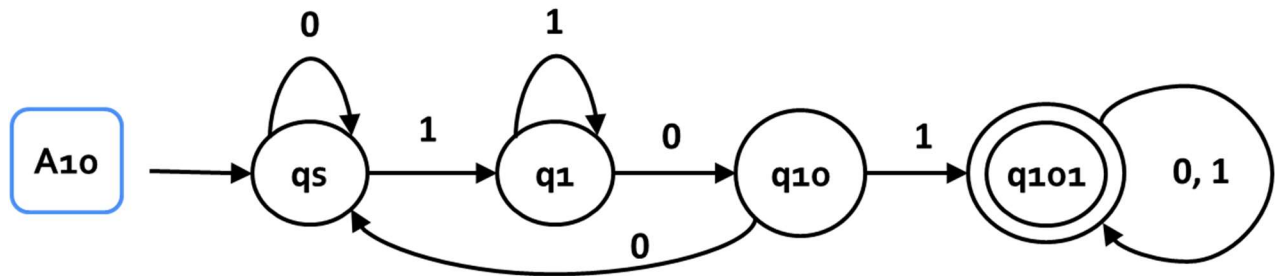


✂ Example 13:

→ $A_{10} = \{w \mid w \text{ has substring } 101\}$. $\Sigma = \{0,1\}$

→ So, $w_1 = 1010$, $w_2 = 001$, $w_3 = 000$, $w_4 = 000110$, $w_5 = 10100...$

Now, $L = \{101, 1101, 10101, 11010, 111010...\}$

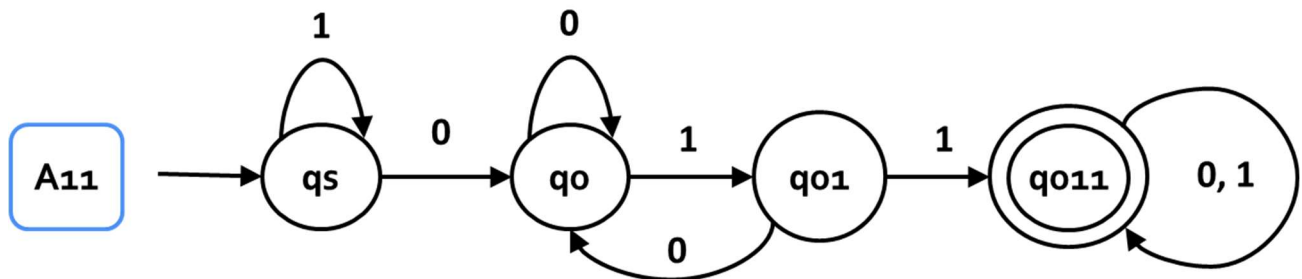


✂ Example 14:

→ $A_{11} = \{w \mid w \text{ has substring } 011\}$. $\Sigma = \{0,1\}$

→ So, $w_1 = 10110$, $w_2 = 001$, $w_3 = 000$, $w_4 = 000110$, $w_5 = 101100...$

Now, $L = \{011, 11011, 10101011, 110110, 1011010...\}$



→ What happens for the language, $A_{11} = \{w \mid w \text{ does not have substring } 011\}$?

→ For $A_{11} = \{w \mid w \text{ has substring "011"}\}$, the DFA **accepts** strings containing "011".

→ For $A'_{11} = \{w \mid w \text{ does not have substring "011"}\}$, the DFA **rejects** any string with "011" and stays in states ensuring "011" never appears.

→ A **dead state** is used for rejection.

Problem 1:

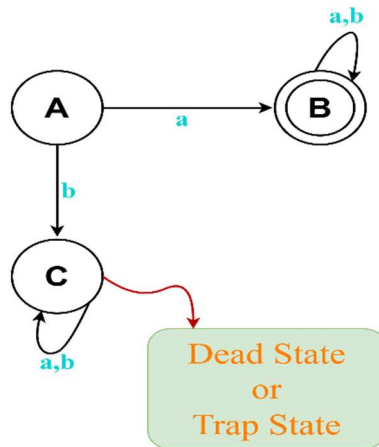
Construct a DFA which accepts set of all string over the $\Sigma = \{a,b\}$, where each string start with an "a"

Or $L(M) = \{w \mid w \text{ Start with an "a"}\}$

→ Is Given,

$\Sigma = \{a,b\}$,

$L(M) = \{a, aa, aba, aabba, \dots\}$



Problem 2:

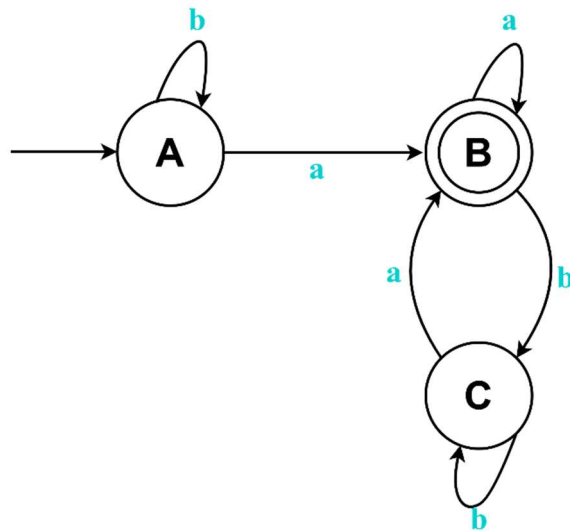
Construct a DFA which accepts set of all string over the $\Sigma = \{a,b\}$, where each string end with an "a"

Or $L(M) = \{w \mid w \text{ End with an "a"}\}$

→ Is Given,

$\Sigma = \{a,b\}$,

$L(M) = \{a, aa, aba, aabba, abababa, bba, \dots\}$



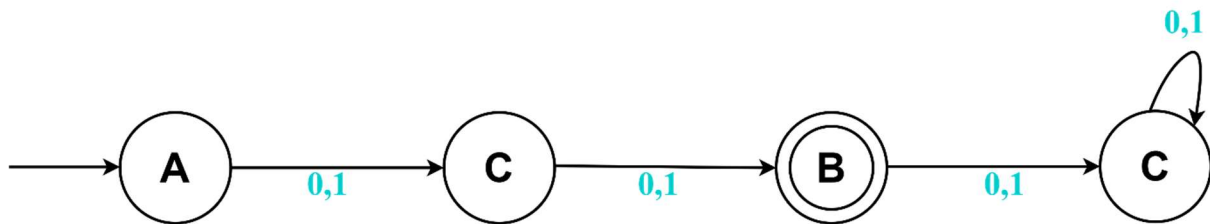
Problem 3:

Construct a DFA that accepts set of all string over the $\Sigma = \{0,1\}$, of length 2.

→ Is Given,

$$\Sigma = \{0,1\},$$

$$L(M) = \{00,01,10,11\}$$



Problem 4:

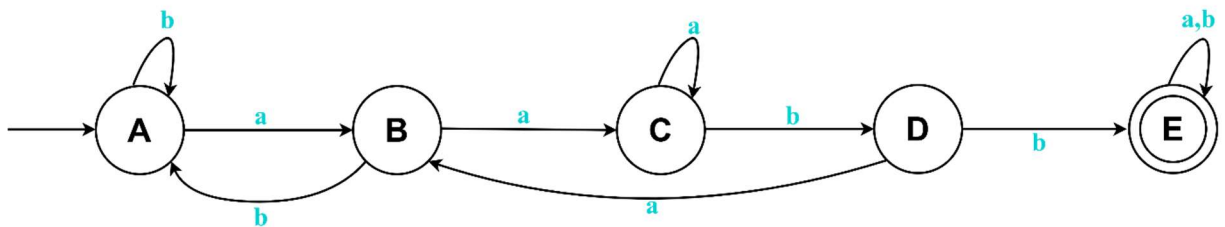
Construct a DFA that accepts any string over the $\Sigma = \{a,b\}$, that does not contain the string "aabb" in it.

→ Is Given,

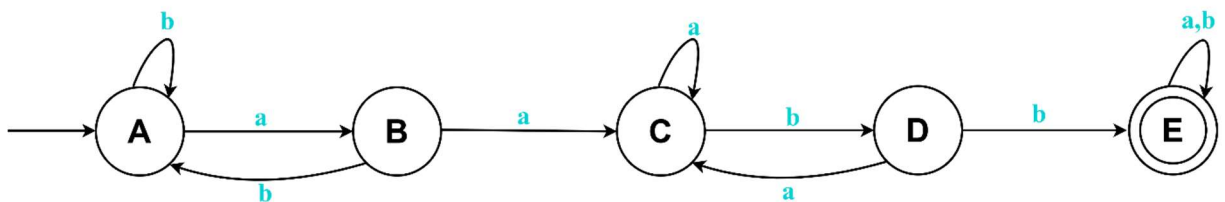
$$\Sigma = \{a,b\},$$

→ Let, Construct a DFA that accepts any string over the $\Sigma = \{a,b\}$, that contains the string "aabb" in it.

$$L(M) = \{aabb, baabb, babaabb, babaaabb, aabaabb, aababaabb, \dots\}$$



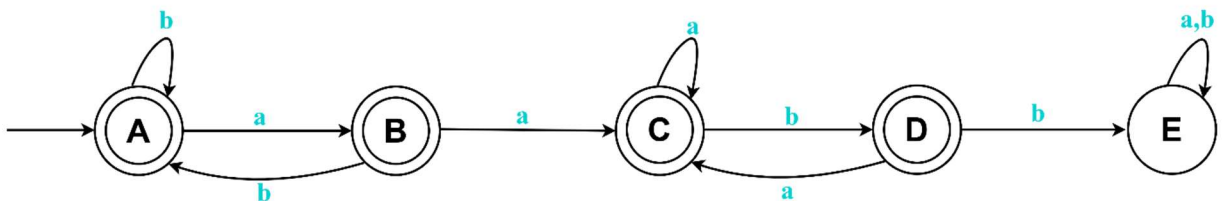
OR



→ Okay, This is not our desired output since it does contain the string "aabb."

→ So, To have our desired output, we just need to:

- Flip the states.
- Make the final states non-final state.
- Make the non-final states final state.



✂ Problem 5:

✂ Design DFA for the following Language. Where $\Sigma = \{a,b\}$ and give the formal definition of your machine. (Any-ONE)

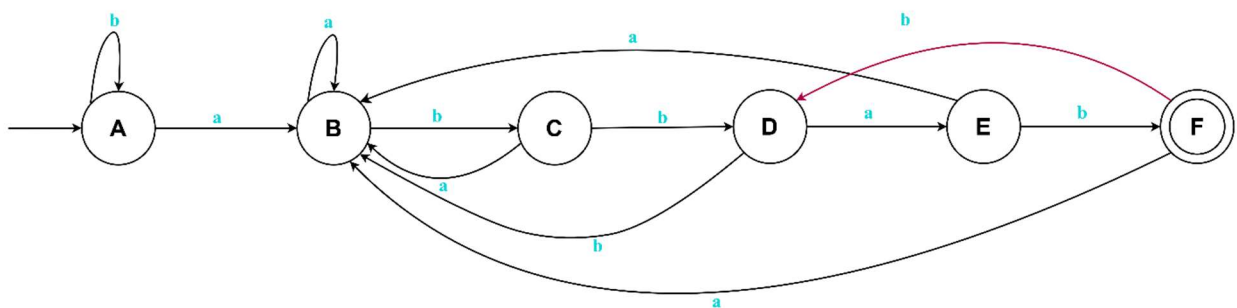
- $L = \{w \mid \text{ends with 'abbab' substring}\}$
- $L = \{w \mid w \text{ contains exactly two a's and odd number of b's}\}$

✂ Mode1#

(i)

→ Is Given,

$\Sigma = \{a,b\}$,
 $L(M1) = \{\text{abbab, babbab, aabbab, ababbab, abbbabbab, abbabbab, abbababbab,}\}$

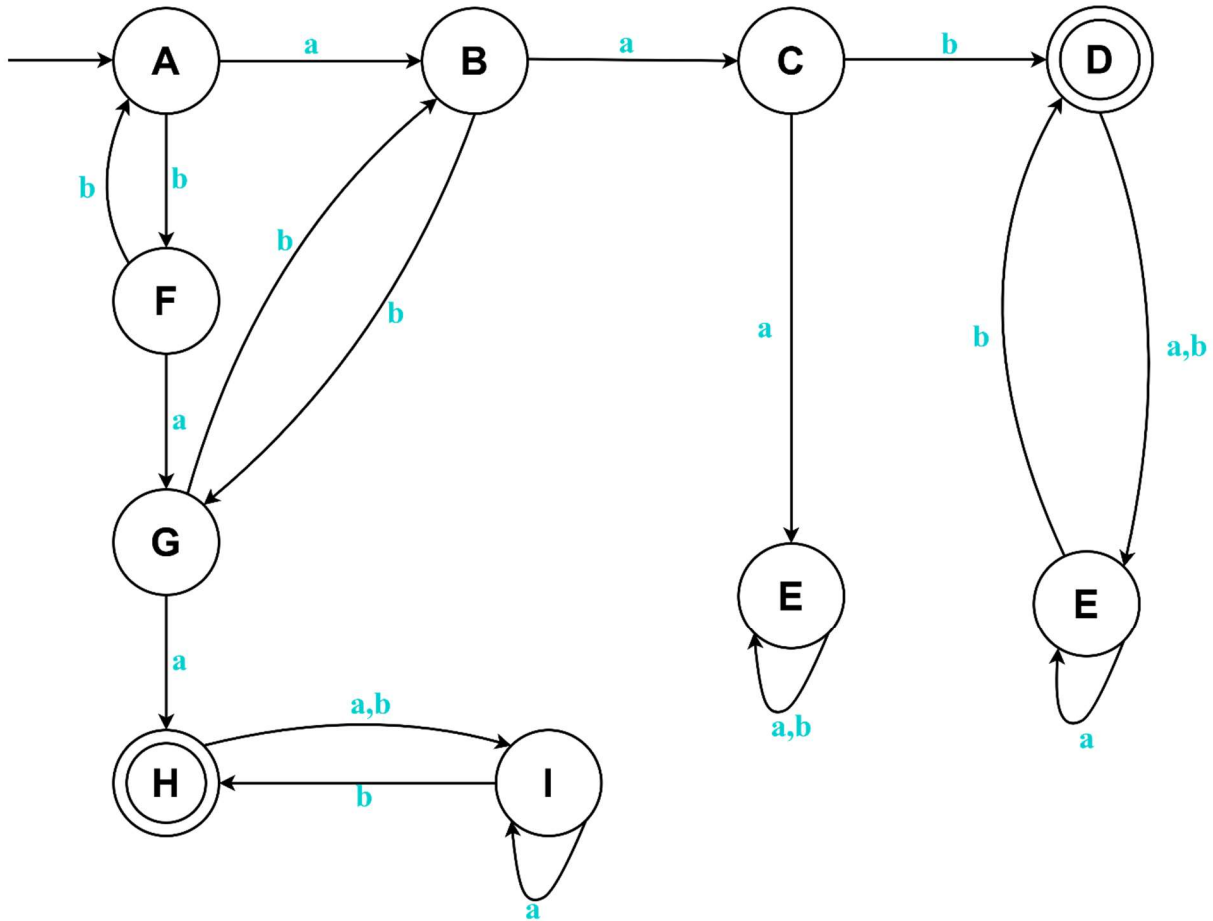


(ii)

→ Is Given,

$\Sigma = \{a,b\}$,

$L(M1) = \{baa, aab, baabb, babab, bbababb, bbababb, abbab, \dots\}$



NON-DETERMINISTIC FINITE AUTOMATA (NFA)

DFA vs NFA

DFA	NFA
» In DFA, given the current state we know what the next state will be » It has only one unique next state » It has no choices or randomness » It is simple and easy to design	» In NFA, given the current state there could be multiple next states. » The next state may be <u>chosen at random</u>. » All the next states may be chosen in parallel » ε-moves are allowed. (ε = empty string)

Running an NFA

➔ NFA Computation:

- Machine **splits** into multiple copies for each choice.
- Each copy follows a different path **in parallel**.
- ε-moves** allow transitions without consuming input.
- Copies **die** if no valid transition exists.
- If **any** copy reaches an **accept state**, the input is accepted.
- Nondeterminism = parallel computation.**

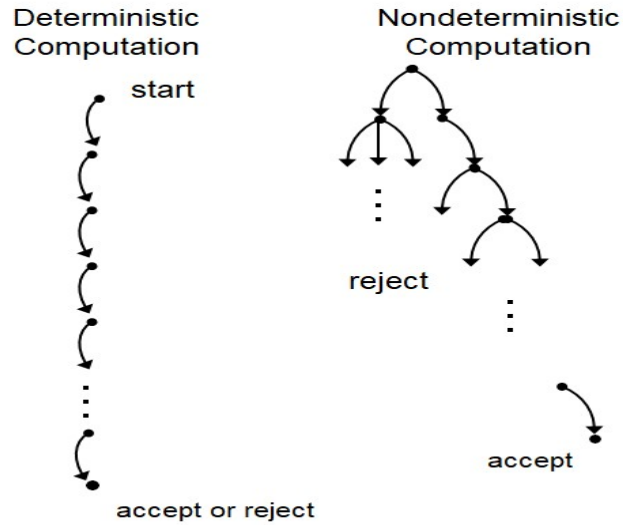
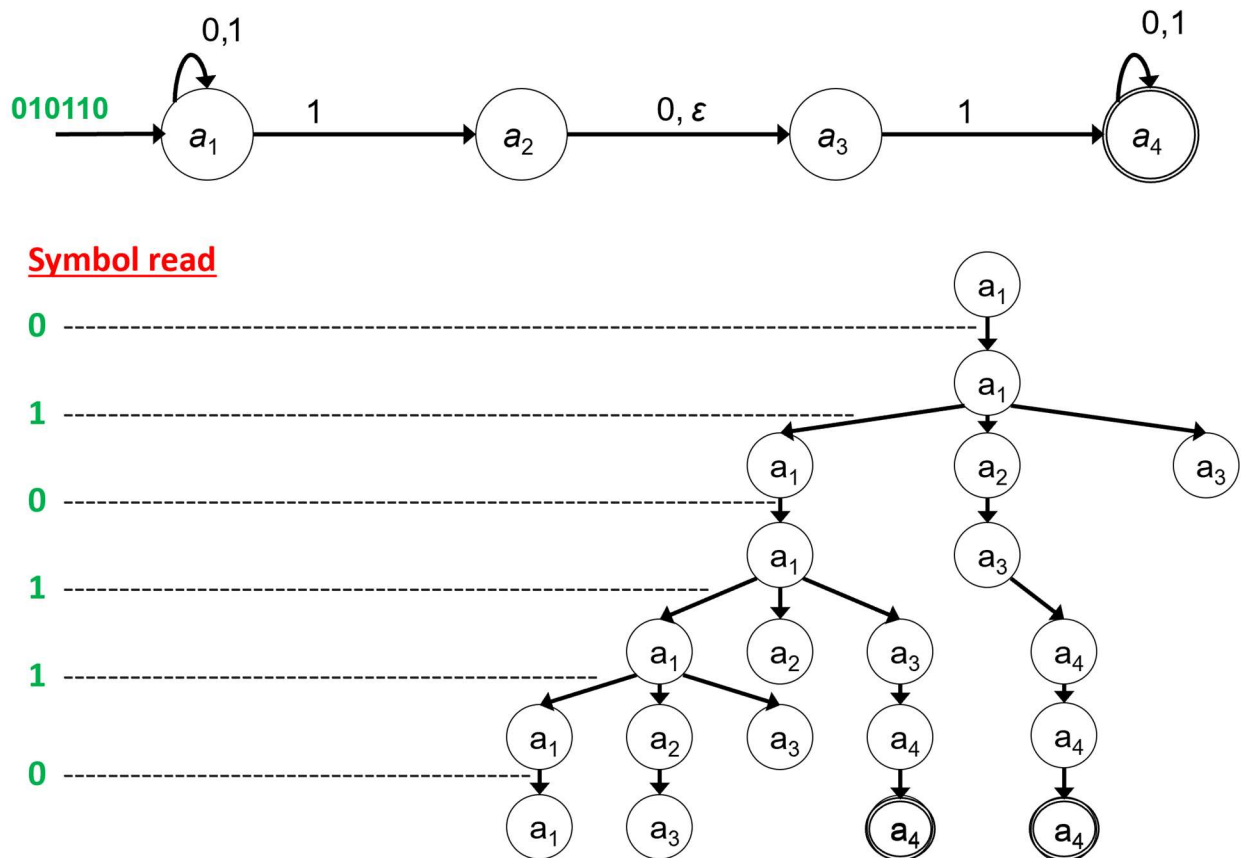


Figure: Deterministic and nondeterministic computations with an accepting branch

Simulation – NFA Tree

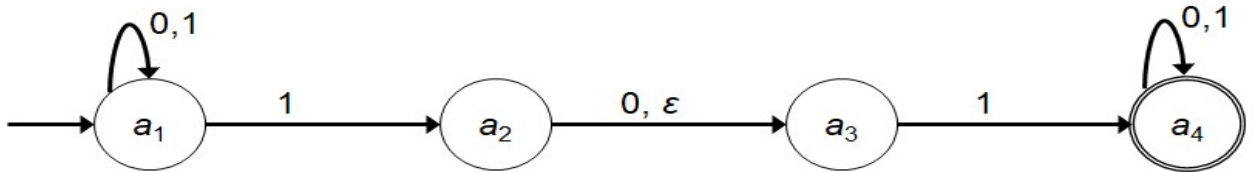


Formal Definition of NFA & Epsilon (ϵ) NFA

➔ NFA as a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$:

- **Q**: Finite set of states.
- **Σ** : Finite alphabet.
- **δ** : Transition function, mapping (state $\times$ symbol/ ϵ) $\rightarrow$ set of states. ($Q \times \Sigma_\epsilon \rightarrow P(Q) = 2^Q$)
- **q_0** : Start state.
- **F**: Set of accepting states.

✧ Example 1:



➔ Here,

» $Q = \{a_1, a_2, a_3, a_4\}$,

» $\Sigma = \{0, 1\}$,

» $q_0 = a_1$,

» $F = \{a_4\}$,

» $\delta =$

δ	0	1	ϵ
a ₁	{a ₁ }	{a ₁ , a ₂ }	ϕ
a ₂	{a ₃ }	ϕ	{a ₃ }
a ₃	ϕ	{a ₄ }	ϕ
a ₄	{a ₄ }	{a ₄ }	ϕ

Fig: Transition table

δ_1 in form $\delta: Q \times \Sigma_\epsilon \rightarrow P(Q)$

$\delta_1(a_1, 0) = \{a_1\}$

$\delta_1(a_1, 1) = \{a_1, a_2\}$

$\delta_1(a_1, \epsilon) = \phi$

$\delta_1(a_2, 0) = \{a_3\}$

$\delta_1(a_2, 1) = \phi$

$\delta_1(a_2, \epsilon) = \{a_3\}$

$\delta_1(a_3, 0) = \phi$

$\delta_1(a_3, 1) = \{a_4\}$

$\delta_1(a_3, \epsilon) = \phi$

$\delta_1(a_4, 0) = \{a_4\}$

$\delta_1(a_4, 1) = \{a_4\}$

$\delta_1(a_4, \epsilon) = \phi$

Fig: Transition Functions

» $Q \times \Sigma_\epsilon = \{(a_1, 0), (a_1, 1), (a_1, \epsilon), (a_2, 0), (a_2, 1), (a_2, \epsilon), (a_3, 0), (a_3, 1), (a_3, \epsilon), (a_4, 0), (a_4, 1), (a_4, \epsilon)\}$

» $P(Q) = \{\phi, \{a_1\}, \{a_2\}, \{a_3\}, \{a_4\}, \{a_1, a_2\}, \{a_1, a_3\}, \{a_1, a_4\}, \{a_2, a_3\}, \{a_2, a_4\}, \{a_3, a_4\}, \{a_1, a_2, a_3\},$

$\{a_1, a_2, a_4\}, \{a_1, a_3, a_4\}, \{a_2, a_3, a_4\}, \{a_1, a_2, a_3, a_4\}\}$

Practice NFA

Problem 1:

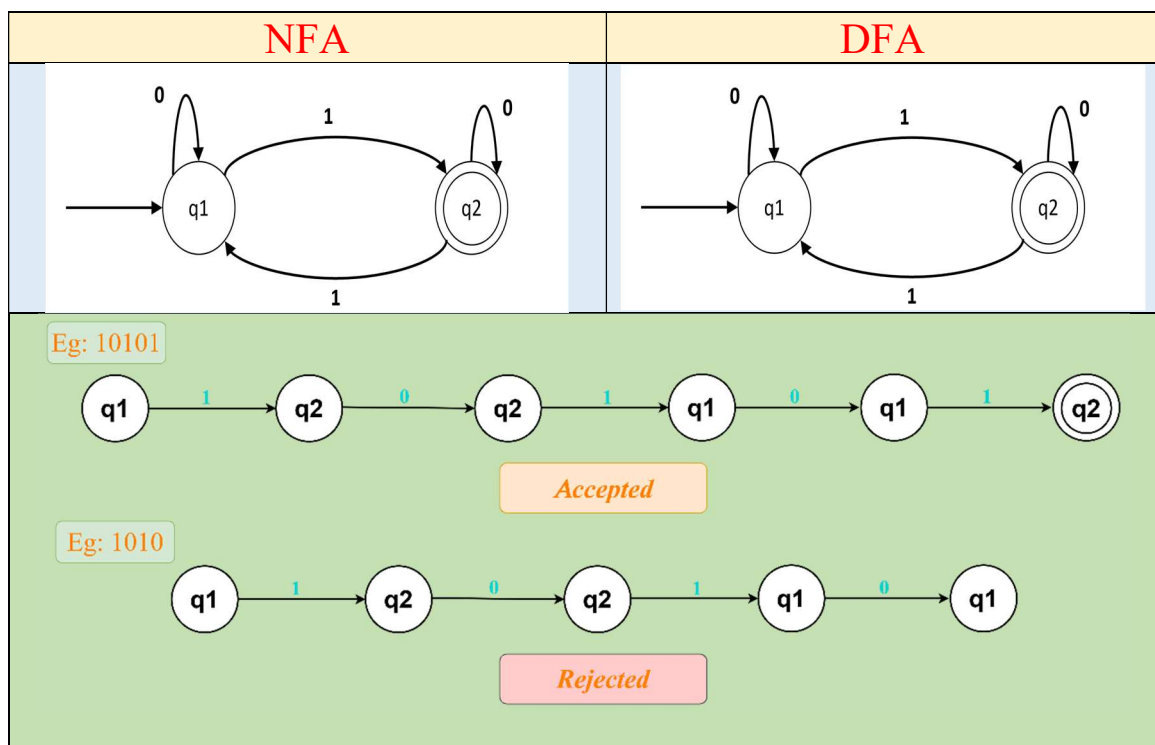
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string containing an **odd number of "1s"**

Or $L(M) = \{w: w \text{ is a binary string containing an odd number of "1s"}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{1, 01, 10, 101, 1101, \dots\}$



Problem 2:

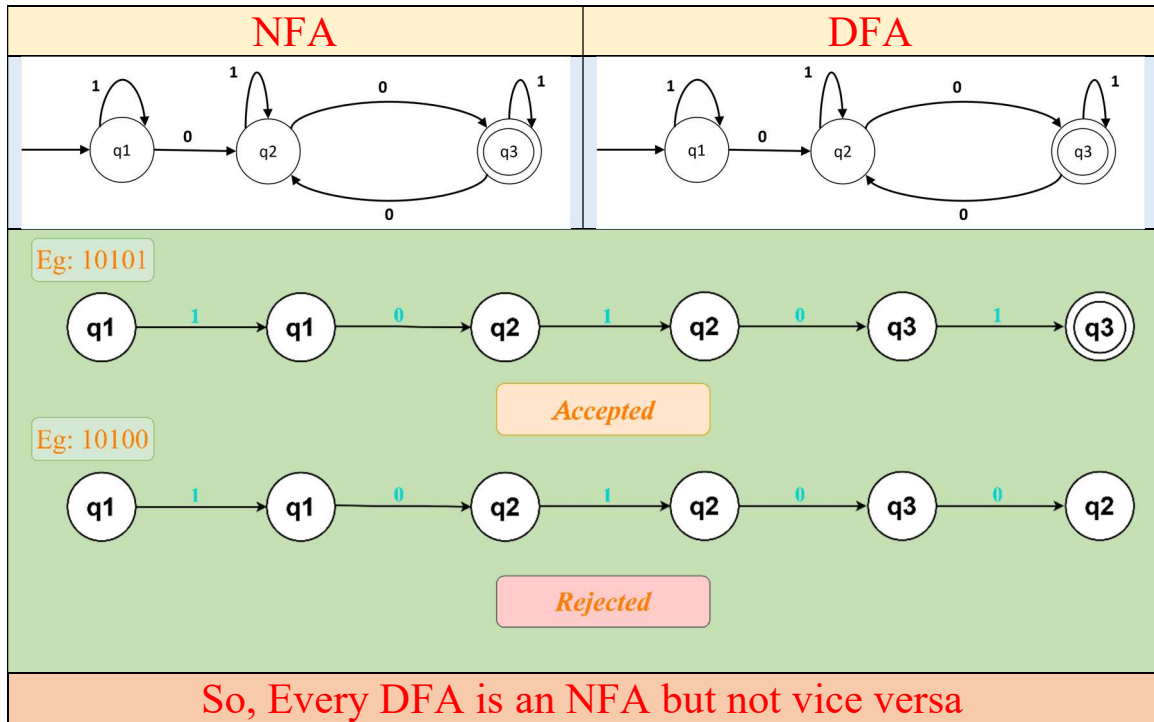
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string containing an **even number of "0s"**

Or $L(M) = \{w: w \text{ is a binary string containing even number of "0s"}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{00, 001, 100, 1001, 11010, \dots\}$



Problem 3:

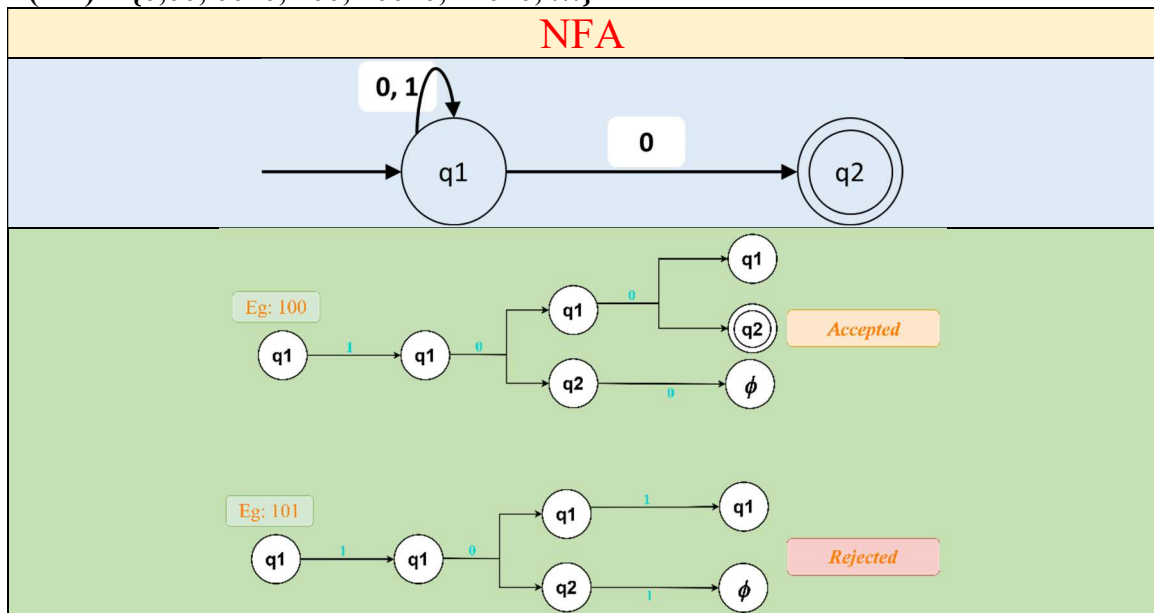
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string ends with a "0"

Or $L(M) = \{w \mid w \text{ ends with a } 0\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{0,00, 0010, 100, 10010, 11010, \dots\}$



🔗 **Problem 4:**

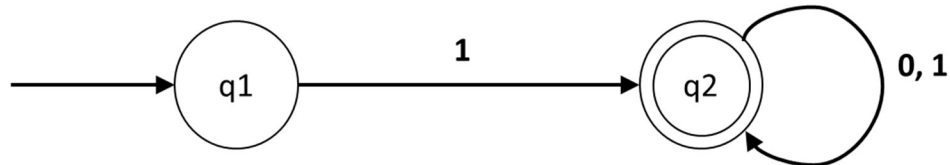
🔗 Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **begins with a "1"**

🔗 Or $L(M) = \{w \mid w \text{ begins with a 1}\}$.

➡ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{1, 10010, 100, 10010, 11010, \dots\}$



🔗 **Problem 5:**

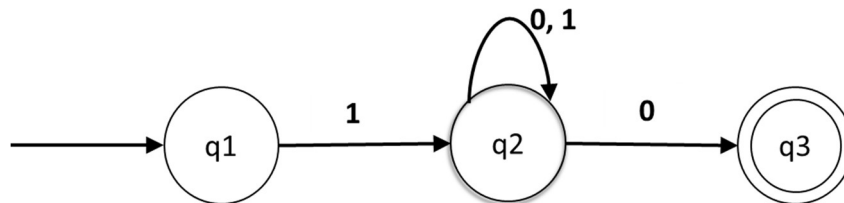
🔗 Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **begins with a 1 and ends with a 0.**

🔗 Or $L(M) = \{w \mid w \text{ begins with a 1 and ends with a 0}\}$.

➡ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{10, 10010, 100, 10010, 11010, \dots\}$



🔗 **Problem 6:**

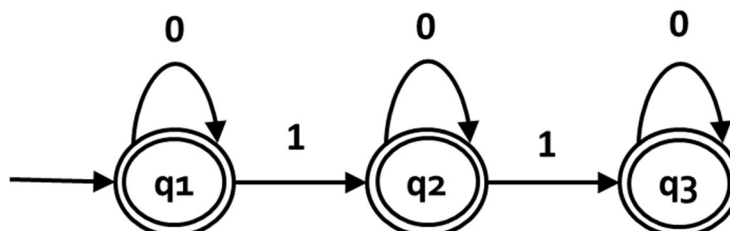
🔗 Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **has at most two 1s**

🔗 Or $L(M) = \{w \mid w \text{ has at most two 1s}\}$.

➡ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{0, 10, 1100, 1010, 10010, 01010, \dots\}$



Problem 7:

Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **has at least two 1s**

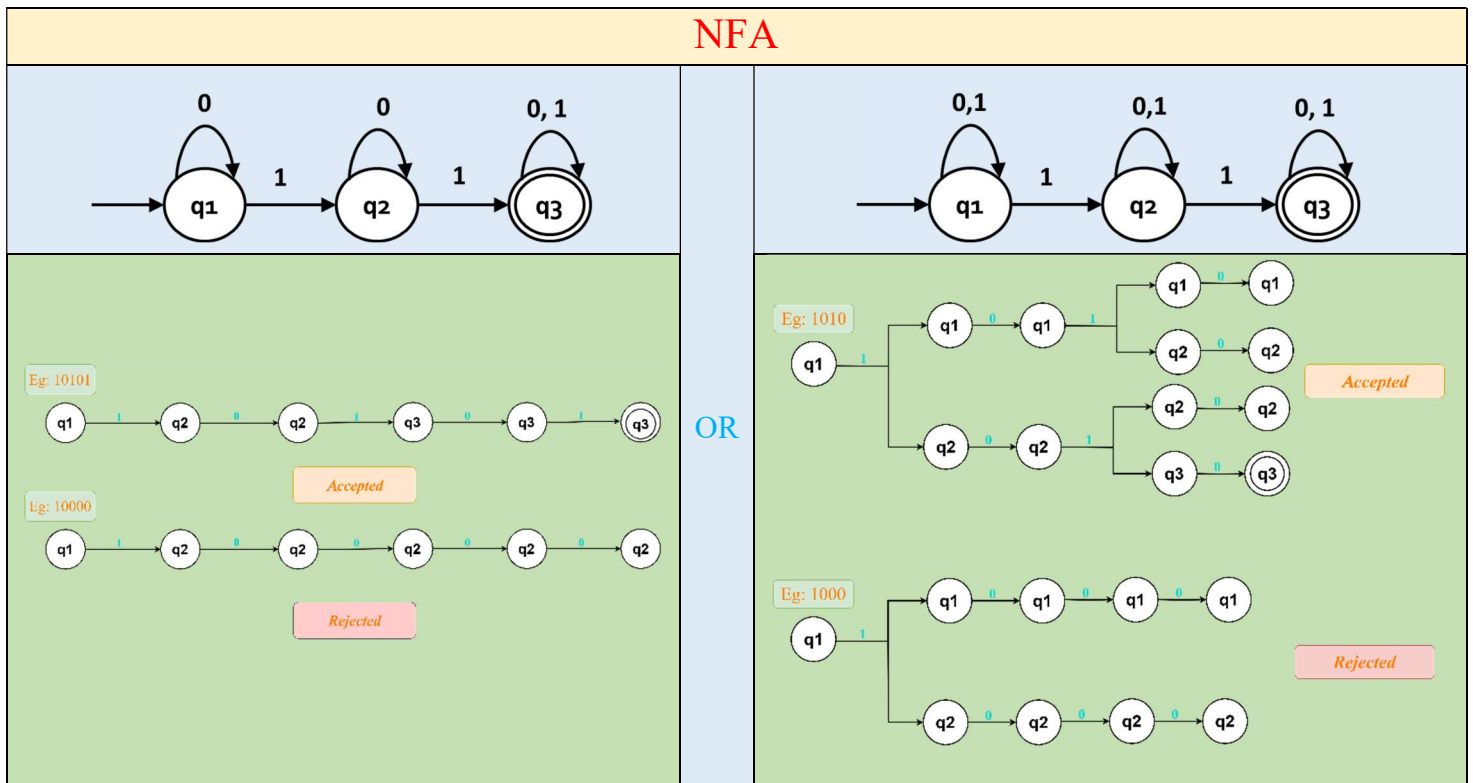
Or $L(M) = \{w \mid w \text{ has at least two 1s}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{11, 110, 01100, 011010, 10010, 01010, \dots\}$

NFA



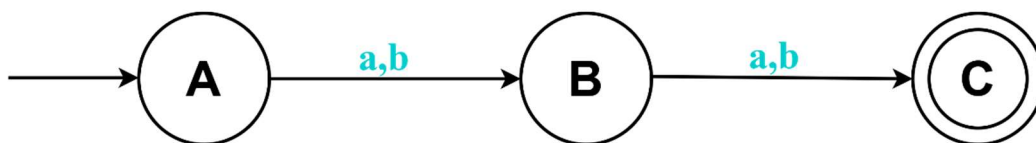
Problem 8:

Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, OF Length 2

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{11, 00, 01, 10\}$



Problem 9:

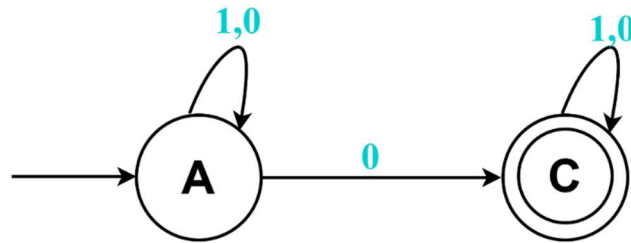
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **Contain "0"**

Or $L(M) = \{w \mid w \text{ Contain "0"}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{0, 010, 011, 10000, \dots\}$



Problem 10:

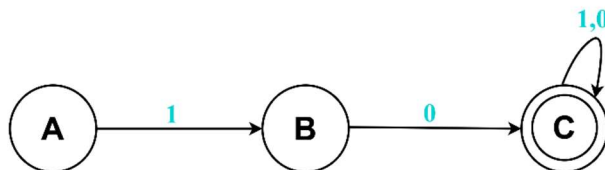
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **begins with "10"**

Or $L(M) = \{w \mid w \text{ begins with "10"}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{10, 1011, 101, 10000, \dots\}$



Problem 11:

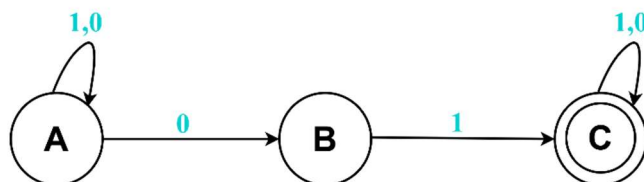
Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **Contain "01"**

Or $L(M) = \{w \mid w \text{ Contain "01"}\}$.

→ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{01, 010, 011, 100001, \dots\}$



✂ **Problem 12:**

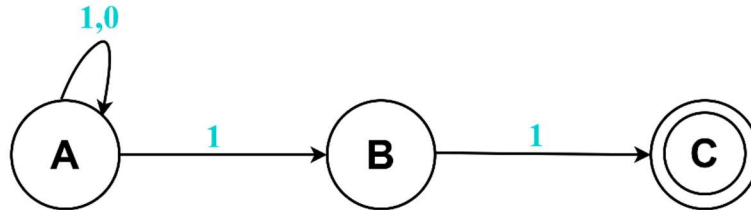
✂ Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **ends with "11"**

✂ Or $L(M) = \{w \mid w \text{ ends with "11"}\}$.

➔ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{11, 011, 0111, 1000011, \dots\}$



✂ **Problem 13:**

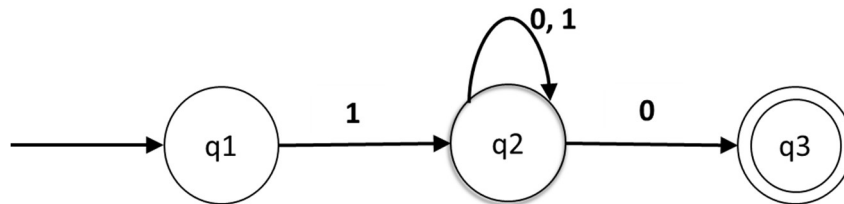
✂ Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **begins with a 1 and ends with a 0.**

✂ Or $L(M) = \{w \mid w \text{ begins with a 1 and ends with a 0}\}$.

➔ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{10, 10010, 100, 10010, 11010, \dots\}$



✂ **Problem 14:**

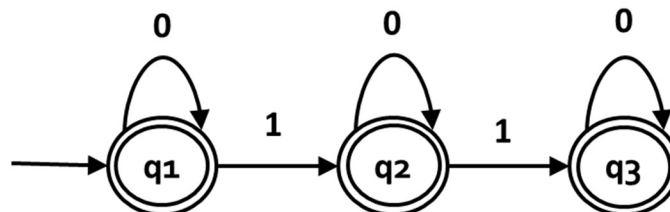
✂ Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **has at most two 1s**

✂ Or $L(M) = \{w \mid w \text{ has at most two 1s}\}$.

➔ Is Given,

$\Sigma = \{0,1\}$,

$L(M) = \{0, 10, 1100, 1010, 10010, 01010, \dots\}$



REGULAR LANGUAGES

Closure

Definition: A collection of objects is *closed* under an operation if applying that operation to its members results in an object still within the collection.

Theorem: The class of **regular languages** is closed under the following operations:

- **Regular operations:** Union, Concatenation, Star
- **Other operations:** Complement, Intersection, Reverse

Regular Languages and Operations

→ A language is **regular** if a finite automaton recognizes it.

Three Basic Regular Operations:

1. **Union ($A \cup B$)** → Combines all strings from both sets.
 - Example: $A = \{\text{good, bad}\}$, $B = \{\text{boy, girl}\}$
 - $A \cup B = \{\text{good, bad, boy, girl}\}$
 - $A \cup B = \{x : x \in A \text{ or } x \in B\}$
2. **Concatenation ($A \cdot B$)** → Attaches each string from A to a string from B.
 - $A \cdot B = \{\text{goodboy, goodgirl, badboy, badgirl}\}$
 - $\{xy : x \in A \text{ and } y \in B\}$
3. **Star (A^*)** → Forms new strings by repeating elements of A any number of times (including 0).
 - $A^* = \{\epsilon, \text{good, bad, goodgood, goodbad, badgood, badbad, ...}\}$
 - ϵ (empty string) is always included.
 - $\{x_1x_2...x_k : k \geq 0 \text{ and each } x_i \in A\}$

Regular Language closed under Union in DFA

→ We construct **M** that recognizes $A_1 \cup A_2$ using **M**₁ and **M**₂:

- **Given:**
 - $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes A_1
 - $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes A_2
- **Constructing M:**
 - **States:** $Q = Q_1 \times Q_2$ (all combinations of states from M_1 and M_2)
 - **Alphabet:** $\Sigma = \Sigma_1 \cup \Sigma_2$ (assumed same for simplicity)
 - **Transition function:**
 - $\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$
 - **Start state:** $q_0 = (q_1, q_2)$
 - **Final states:** $F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$ (if either machine reaches an accept state)

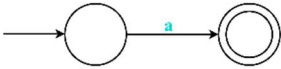
Thus, **M** recognizes $A_1 \cup A_2$ by simulating both M_1 and M_2 in parallel.

Closure under Concatenation

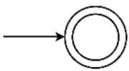
- **Problem:** M must determine where to split the input for M_1 and M_2 .
- **Solution:** Use a **nondeterministic automaton (NFA)** to guess the split point.
- **Construction of $M = (Q, \Sigma, \delta, q_0, F)$:**
 - **States:** $Q = Q_1 \times Q_2 \cup \{q_0\}$
 - **Start State:** q_0 (new initial state)
 - **Transition Function:**
 - Simulate M_1 on the input.
 - At any step, nondeterministically transition to M_2 if M_1 reaches an accepting state.
 - Simulate M_2 from that point onward.
 - **Final States:** $F = \{(q_1, q_2) \mid q_1 \in F_1 \text{ and } q_2 \in F_2\}$
- **Key Idea:** M nondeterministically decides where M_1 ends and M_2 begins, ensuring acceptance if such a split exists.

Formal Definition of Regular Language

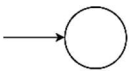
1. $R = a$; $a \in \Sigma$



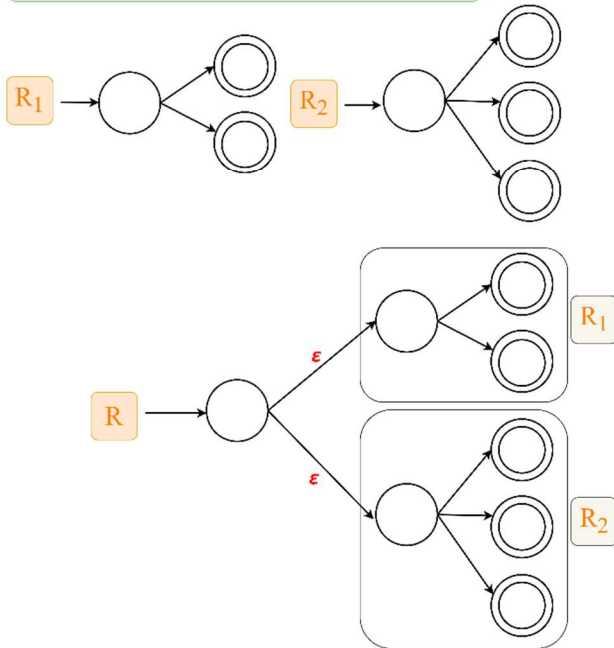
2. $R = \epsilon$; initial state is the final state



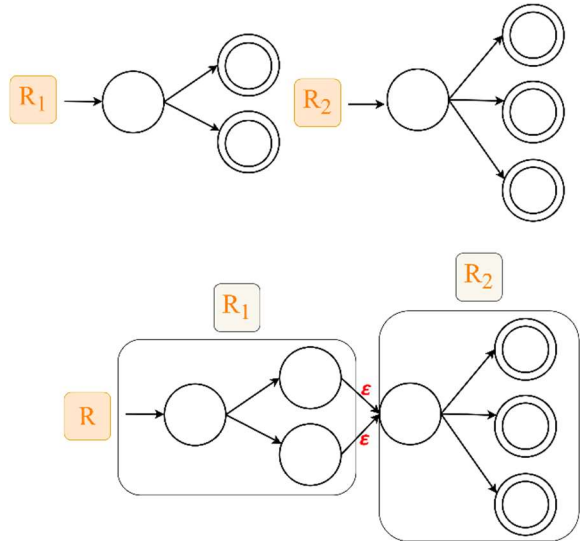
3. $R = \phi$



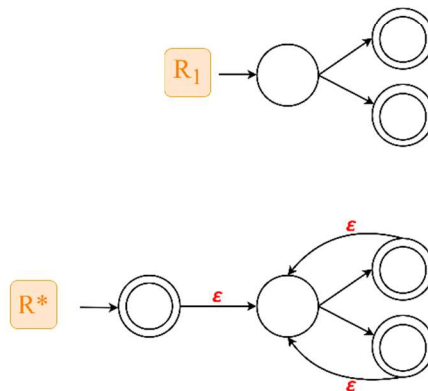
4. $R = R_1 \cup R_2$ or $R = R_1 | R_2$ or $R = R_1$ or R_2



5. $R = R_1 R_2$ or $R = R_1 R_2$ or $R = R_1$ Followed by R_2



6. $R^* = R_1^*$



Practice Problem

✂ Example 1:

if $B_6 = \{w \mid w \text{ has at least two 1s}\}.$

$B_7 = \{w \mid w \text{ has at most two 1s}\}.$

$B_{11} = \{w \mid w \text{ has at least two 1s or } w \text{ has at most two 1s}\}.$

Where, $\Sigma = \{0,1\}$

Then Draw the NFA machine.

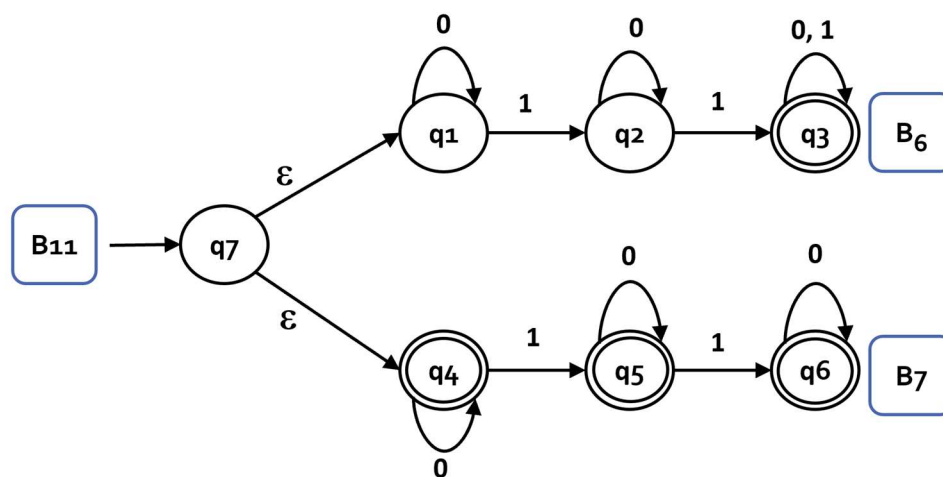
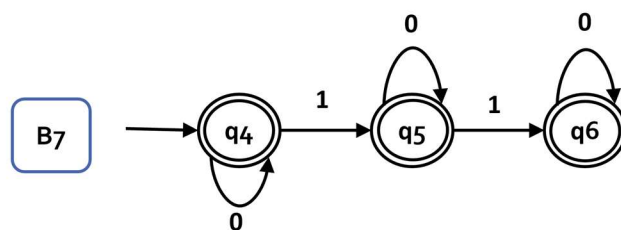
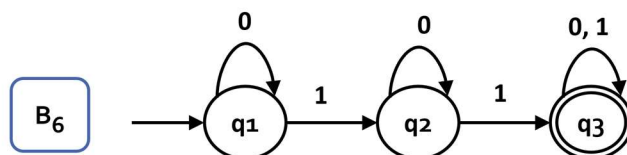
➔ Is Given,

$\Sigma = \{0,1\},$

$L(B_6) = \{1, 01, 10, 1100, 1010, 10010, 01010, \dots\}$

$L(B_7) = \{1, 01, 10, 1100, 1010, 10010, 01010, \dots\}$

$L(B_{11}) = (L(B_6) \cup L(B_7))$



∅ **Example 2:**

if $B_6 = \{w \mid w \text{ has at least two } 1s\}$.

$B_7 = \{w \mid w \text{ has at most two } 1s\}$.

$B_{12} = \{w \mid w \text{ has at least two } 1s \text{ followed by } w \text{ has at most two } 1s\}$.

Where, $\Sigma = \{0,1\}$

Then Draw the NFA machine.

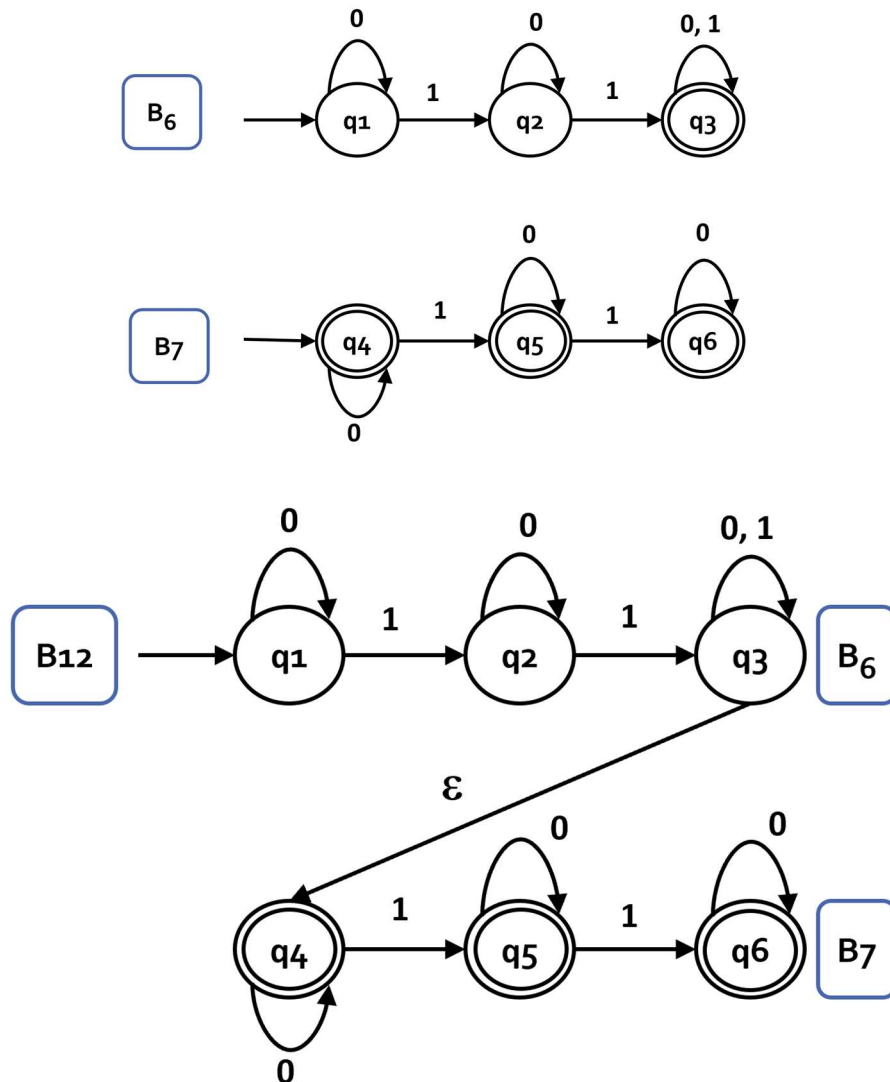
→ Is Given,

$\Sigma = \{0,1\}$,

$L(B_6) = \{1, 01, 10, 1100, 1010, 10010, 01010, \dots\}$

$L(B_7) = \{1, 01, 10, 1100, 1010, 10010, 01010, \dots\}$

$L(B_{12}) = (L(B_6) \circ L(B_7))$



☞ **Example 3:**

if $NFA\ B_{10} = \{w \mid w \text{ starts with a 1 and has even length or } w \text{ starts with a 0 and has odd length}\}.$

Find B_{10}^* for $\Sigma = \{0,1\}$

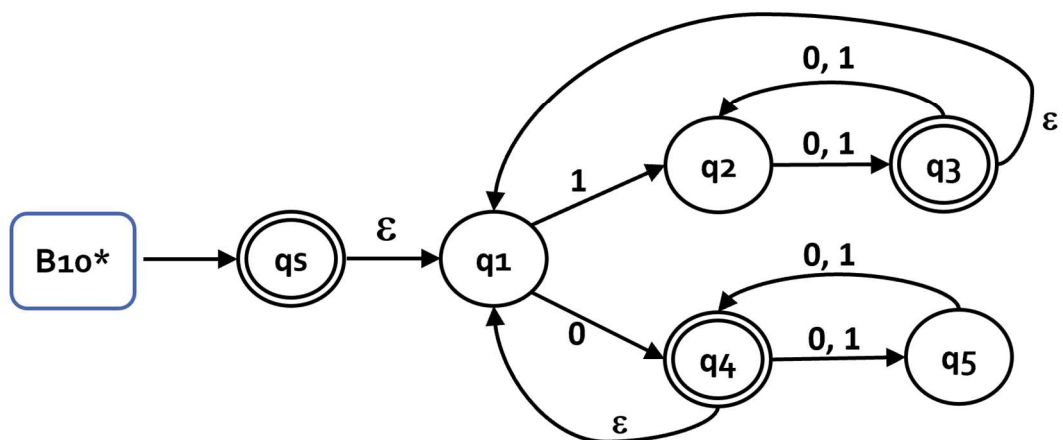
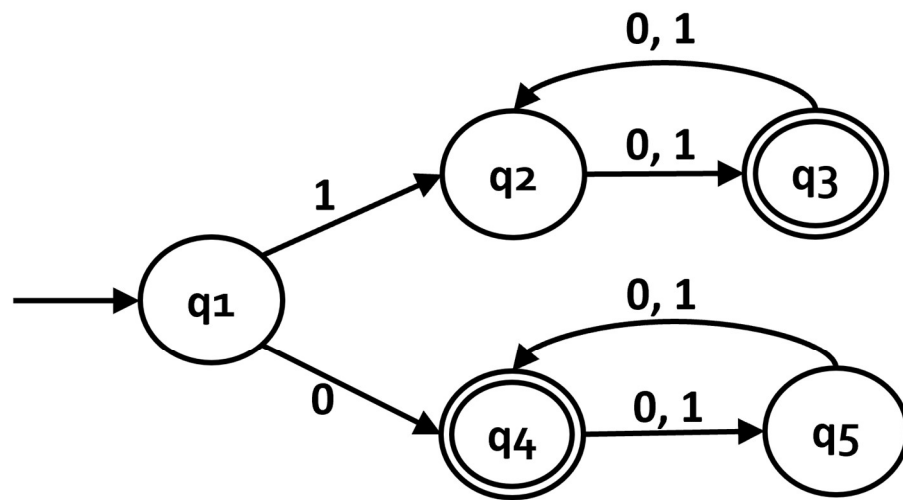
➔ **Is Given,**

$\Sigma = \{0,1\},$

$L(M) = \{11, 10, 1000, 1100, 1010, 101110, 1100, \dots\}$

$L(M) = \{0, 010, 01110, 01100, 01010, 00010, 01010, \dots\}$

$L(B_{10})^* = (L(B_6) \text{ or } L(B_7))^*$

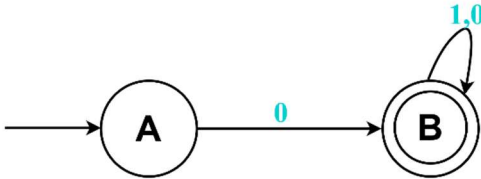
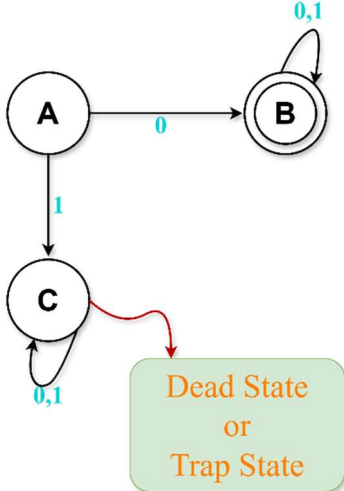


CONVERSION OF NFA TO DFA

Conversion of NFA to DFA

➤ Every DFA is an NFA, but not vice versa, But there is an **equivalent** DFA for every NFA

✂ **Example1:** Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **Start with a "0"**. Construct the **equivalent** DFA.

Step1: Construct the NFA	Step2: Write the <u>transition table</u> of NFA												
	<table><tr><th>δ</th><th>0</th><th>1</th></tr><tr><td>→ A</td><td>{B}</td><td>ϕ</td></tr><tr><td>ⓑ</td><td>{B}</td><td>{B}</td></tr></table>	δ	0	1	→ A	{B}	ϕ	ⓑ	{B}	{B}			
δ	0	1											
→ A	{B}	ϕ											
ⓑ	{B}	{B}											
Step4: Now, Construct the DFA from T.table	Step3: Convert this NFA transition table to its equivalent DFA <u>transition table</u> .												
	<table><tr><th>δ</th><th>0</th><th>1</th></tr><tr><td>→ A</td><td>{B}</td><td>{C}</td></tr><tr><td>ⓑ</td><td>{B}</td><td>{B}</td></tr><tr><td>ⓒ</td><td>{C}</td><td>{C}</td></tr></table>	δ	0	1	→ A	{B}	{C}	ⓑ	{B}	{B}	ⓒ	{C}	{C}
δ	0	1											
→ A	{B}	{C}											
ⓑ	{B}	{B}											
ⓒ	{C}	{C}											

Practice Problem

✂ **Problem1:** Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **End with a “1”**. Also Construct the **equivalent DFA**.

Step1: Construct the NFA	Step2: Write the <u>transition table</u> of NFA	Explanations									
	<table border="1"> <thead> <tr> <th>δ</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <td>→ A</td><td>{A}</td><td>{A, B}</td></tr> <tr> <td>ⓑ</td><td>ϕ</td><td>ϕ</td></tr> </tbody> </table>	δ	0	1	→ A	{A}	{A, B}	ⓑ	ϕ	ϕ	
δ	0	1									
→ A	{A}	{A, B}									
ⓑ	ϕ	ϕ									
Step4: Now, Construct the DFA from T.table	Step3: Convert this NFA transition table to its equivalent <u>DFA transition table</u> .										
	<table border="1"> <thead> <tr> <th>δ</th><th>0</th><th>1</th></tr> </thead> <tbody> <tr> <td>→ A</td><td>{A}</td><td>{AB}</td></tr> <tr> <td>ⓐ</td><td>{A}</td><td>{AB}</td></tr> </tbody> </table>	δ	0	1	→ A	{A}	{AB}	ⓐ	{A}	{AB}	AB = new Single State $A \xrightarrow{1} (AUB) = AB$ $A \xrightarrow{0} (AUB) = A$
δ	0	1									
→ A	{A}	{AB}									
ⓐ	{A}	{AB}									

✂ **Problem2:** Find the equivalent DFA for the NFA given by $M = [\{A, B, C\}, (a,b), \delta, A, \{C\}]$ where δ is given by:

δ	a	b
A	{A, B}	{C}
B	{A}	{B}
C	-	{A, B}

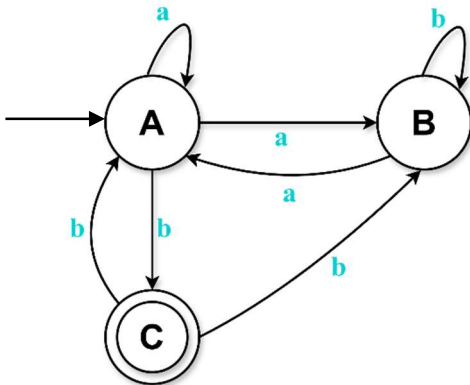
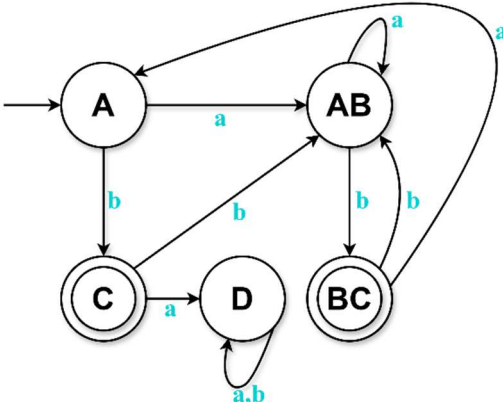
➔ **Here,**

» $Q = \{A, B, C\}$,

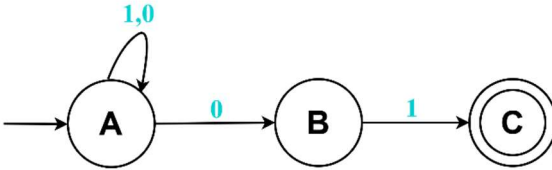
» $\Sigma = \{a, b\}$,

» $q_0 = A$,

» $F = \{C\}$,

Step1: Construct the NFA	Step2: Write the <u>transition table</u> of NFA																		
	<table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow$ A</td><td>{A, B}</td><td>{C}</td></tr><tr><td>B</td><td>{A}</td><td>{B}</td></tr><tr><td>C</td><td>ϕ</td><td>{A, B}</td></tr></table>	δ	a	b	$\rightarrow$ A	{A, B}	{C}	B	{A}	{B}	C	ϕ	{A, B}						
δ	a	b																	
$\rightarrow$ A	{A, B}	{C}																	
B	{A}	{B}																	
C	ϕ	{A, B}																	
Step4: Now, Construct the DFA from T.table	Step3: Convert this NFA transition table to its equivalent <u>DFA transition table</u> .																		
	<table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow$ A</td><td>{AB}</td><td>{C}</td></tr><tr><td>AB</td><td>{AB}</td><td>{BC}</td></tr><tr><td>BC</td><td>{A}</td><td>{AB}</td></tr><tr><td>C</td><td>{D}</td><td>{AB}</td></tr><tr><td>D</td><td>{D}</td><td>{D}</td></tr></table>	δ	a	b	$\rightarrow$ A	{AB}	{C}	AB	{AB}	{BC}	BC	{A}	{AB}	C	{D}	{AB}	D	{D}	{D}
δ	a	b																	
$\rightarrow$ A	{AB}	{C}																	
AB	{AB}	{BC}																	
BC	{A}	{AB}																	
C	{D}	{AB}																	
D	{D}	{D}																	

✂ **Problem3:** Construct a NFA which accepts set of all string over the $\Sigma = \{0,1\}$, where a binary string **Ends with a “01”**. Also Construct the **equivalent DFA**.

Step1: Construct the NFA	Step2: Write the <u>transition table</u> of NFA												
	<table><tr><th>δ</th><th>0</th><th>1</th></tr><tr><td>$\rightarrow$A</td><td>{A, B}</td><td>{A}</td></tr><tr><td>B</td><td>ϕ</td><td>{C}</td></tr><tr><td>C</td><td>ϕ</td><td>ϕ</td></tr></table>	δ	0	1	$\rightarrow$ A	{A, B}	{A}	B	ϕ	{C}	C	ϕ	ϕ
δ	0	1											
$\rightarrow$ A	{A, B}	{A}											
B	ϕ	{C}											
C	ϕ	ϕ											

Epsilon (ϵ) NFA to DFA conversion

✂ Example1: Find the equivalent DFA for the NFA given by $M = [\{b_1, b_2, b_3\}, (0,1), \delta, b_1, \{b_1\}]$ where δ is given by:

δ	0	1	ϵ
b_1	ϕ	$\{b_2\}$	$\{b_3\}$
b_2	$\{b_2, b_3\}$	$\{b_3\}$	ϕ
b_3	$\{b_1\}$	ϕ	ϕ

➔ Here,

» $Q = \{b_1, b_2, b_3\},$

» $\Sigma = \{0, 1\},$

» $q_0 = b_1,$

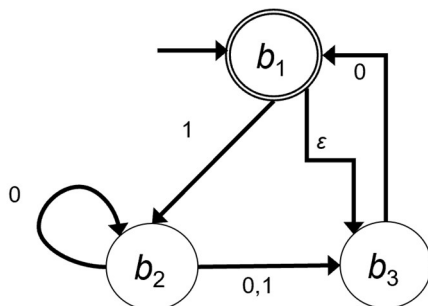
» $F = \{b_1\},$

➔ For Equivalent DFA:

» $Q = P(Q) = P(\{b_1, b_2, b_3\})$

» $P(Q) = \{ \phi, \{b_1\}, \{b_2\}, \{b_3\}, \{b_1, b_2\}, \{b_1, b_3\}, \{b_2, b_3\}, \{b_1, b_2, b_3\} \}$

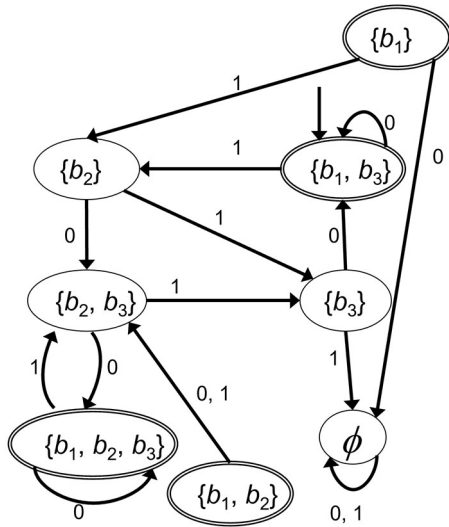
Step1: Construct the NFA



Step2: Write the transition table of NFA

δ	0	1	ϵ
b_1	ϕ	$\{b_2\}$	$\{b_3\}$
b_2	$\{b_2, b_3\}$	$\{b_3\}$	ϕ
b_3	$\{b_1\}$	ϕ	ϕ

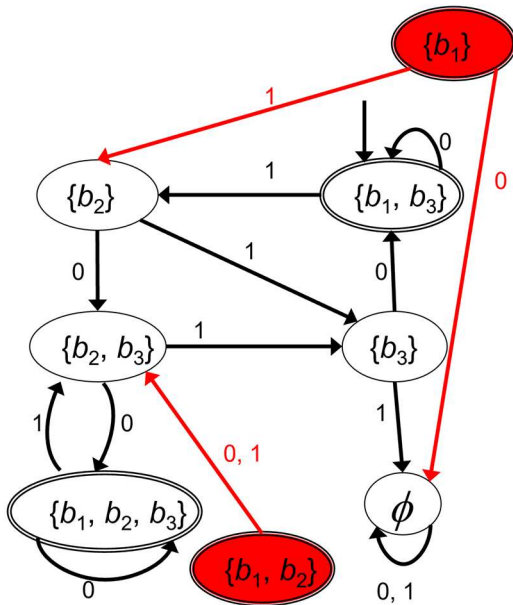
Step4: Now, Construct the DFA from T.table



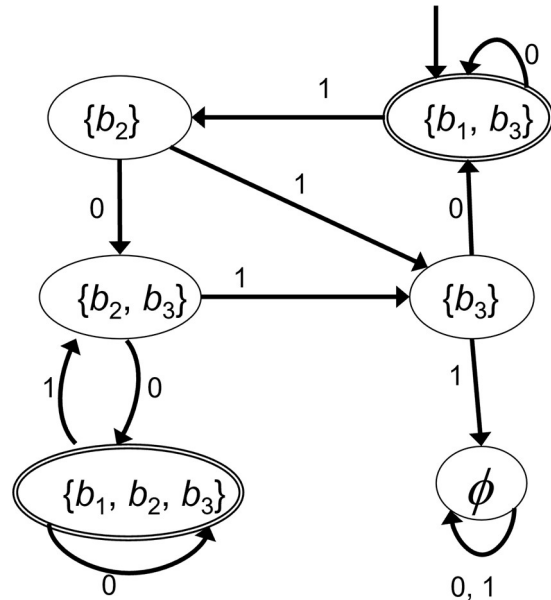
Step3: Convert this NFA transition table to its equivalent **DFA** transition table.

δ	0	1
ϕ	ϕ	ϕ
b_1	ϕ	$\{b_2\}$
b_2	$\{b_2, b_3\}$	$\{b_3\}$
b_3	$\{b_1, b_3\}$	ϕ
$\{b_1, b_2\}$	$\{b_2, b_3\}$	$\{b_2, b_3\}$
$\{b_1, b_3\}$	$\{b_1, b_3\}$	$\{b_2\}$
$\{b_2, b_3\}$	$\{b_1, b_2, b_3\}$	$\{b_3\}$
$\{b_1, b_2, b_3\}$	$\{b_1, b_2, b_3\}$	$\{b_2, b_3\}$

Step5: Remove the states with no incoming arrows.

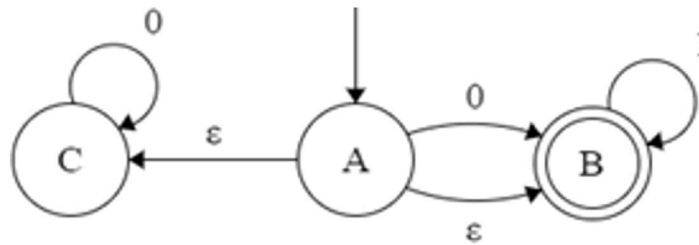


Step6: After Remove the states with no incoming arrows.



Practice Problem

✂ **Problem1:** Convert the following NFA into equivalent DFA using the formal procedure.



➔ **Here,**

» $Q = \{A, B, C\}$,

» $\Sigma = \{0, 1\}$,

» $q_0 = A$,

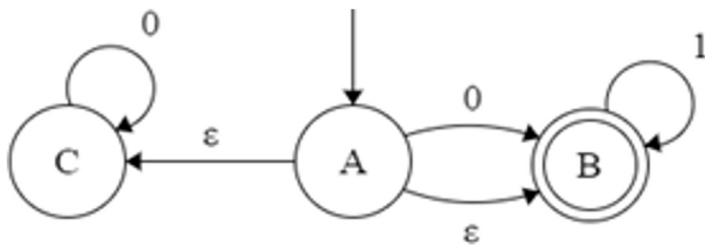
» $F = \{B\}$,

➔ **For Equivalent DFA:**

» $Q = P(Q) = P(\{A, B, C\})$

» $\mathcal{P}(Q) = \{ \phi, \{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A, B, C\} \}$

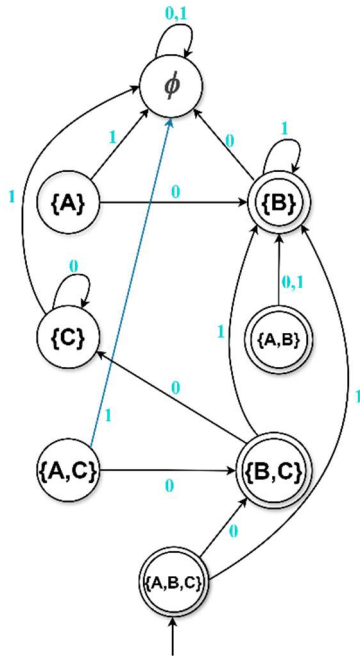
Step1: Construct the NFA



Step2: Write the transition table of NFA

δ	0	1	ϵ
A	{B}	ϕ	{B, C}
<u>B</u>	ϕ	{B}	ϕ
C	{C}	ϕ	ϕ

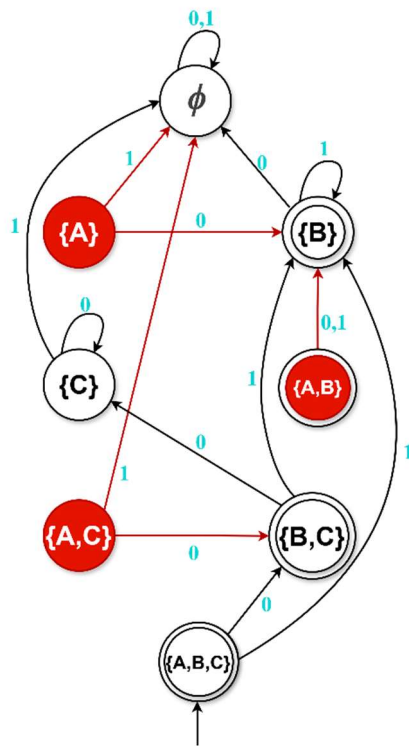
Step4: Now, Construct the DFA from T.table



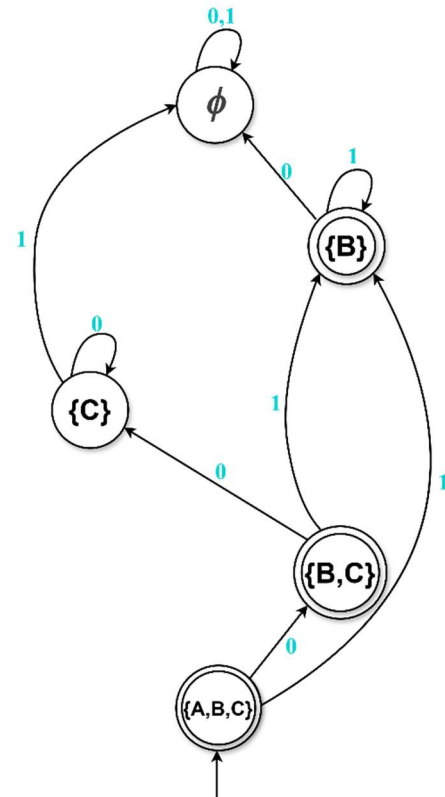
Step3: Convert this NFA transition table to its equivalent **DFA** transition table.

δ	0	1
$\emptyset$	$\emptyset$	$\emptyset$
$\{A\}$	$\{B\}$	$\emptyset$
$\{B\}$	$\emptyset$	$\{B\}$
$\{C\}$	$\{C\}$	$\emptyset$
$\{A, B\}$	$\{B\}$	$\{B\}$
$\{A, C\}$	$\{B, C\}$	$\emptyset$
$\{B, C\}$	$\{C\}$	$\{B\}$
$\{A, B, C\}$	$\{B, C\}$	$\{B\}$

Step5: Remove the states with no incoming arrows.

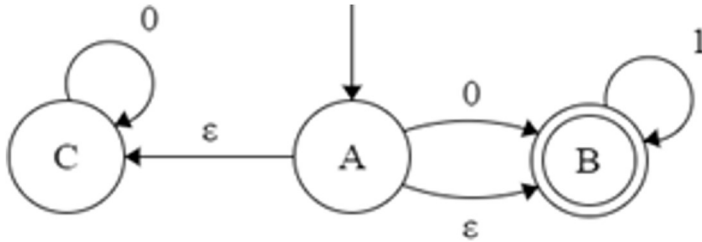


Step6: After Remove the states with no incoming arrows.



"An Alternative Approach to Solving It"

Step1: Construct the NFA



Step2: Write the transition table of NFA

δ	0	1	ϵ
A	{B}	ϕ	{B, C}
B	ϕ	{B}	ϕ
C	{C}	ϕ	ϕ

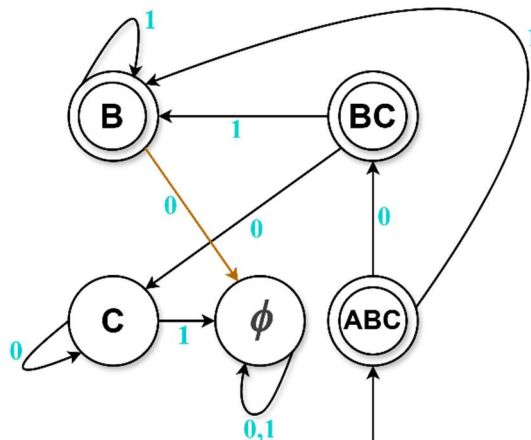
Step3: Find the Epsilon (ϵ) Closure of all the state

- ϵ -Closure for {A} = {A, B, C}
- ϵ -Closure for {B} = {B}
- ϵ -Closure for {C} = {C}

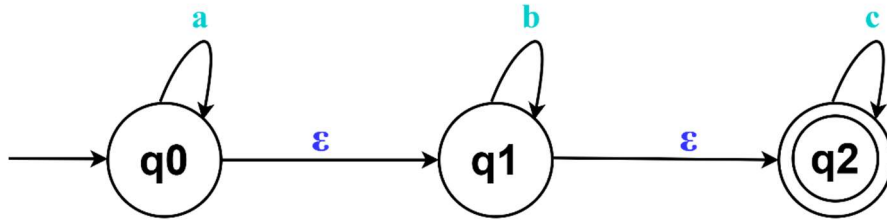
Step4: Convert this NFA transition table to its equivalent DFA transition table.

δ	0	1
{A, B, C}	{B, C}	{B}
{B, C}	{C}	{B}
{B}	ϕ	{B}
{C}	{C}	ϕ
ϕ	ϕ	ϕ

Step5: Now, Construct the DFA from transition table.



🔗 **Problem2:** Convert the following NFA into equivalent DFA using the formal procedure.



➔ **Here,**

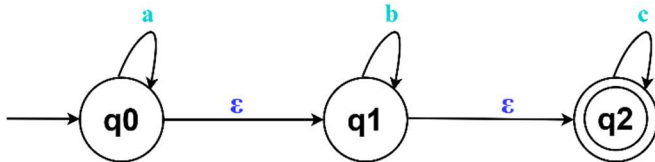
» $Q = \{q_0, q_1, q_2\}$,

» $\Sigma = \{a, b\}$,

» $q_0 = q_0$,

» $F = \{q_2\}$,

Step1: Construct the NFA



Step2: Write the transition table of NFA

δ	a	b	c	ϵ
q_0	$\{q_0\}$	ϕ	ϕ	$\{q_1\}$
q_1	ϕ	$\{q_1\}$	ϕ	$\{q_2\}$
q_2	ϕ	ϕ	$\{q_2\}$	ϕ

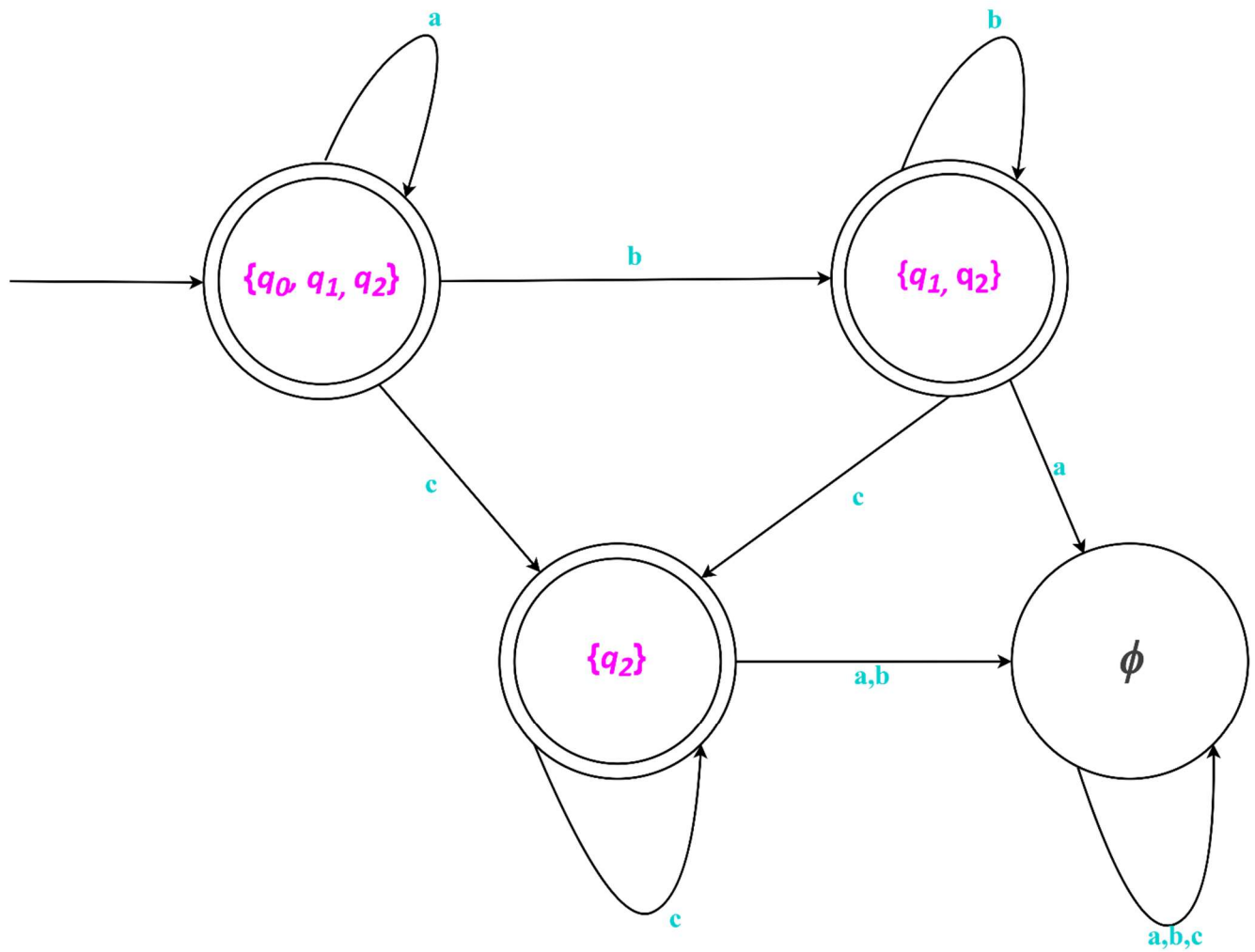
Step3: Find the Epsilon (ϵ) Closure of all the state

- ϵ -Closure for $\{q_0\} = \{q_0, q_1, q_2\}$
- ϵ -Closure for $\{q_1\} = \{q_1, q_2\}$
- ϵ -Closure for $\{q_2\} = \{q_2\}$

Step4: Convert this NFA transition table to its equivalent **DFA** transition table.

δ	a	b	c
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_1, q_2\}$	ϕ	$\{q_1, q_2\}$	$\{q_2\}$
$\{q_2\}$	ϕ	ϕ	$\{q_2\}$
ϕ	ϕ	ϕ	ϕ

Step5: Now, Construct the DFA from transition table.



REGULAR EXPRESSION

Regular Expression

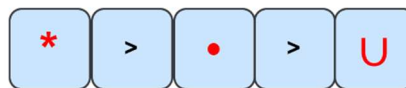
>>Regular expressions are used to represent certain sets of strings in an **algebraic** fashion, moving beyond simple English representations of strings.

- Structure:

Built using alphabets and **regular operations**:

- * (star/**Closure**) – Represents zero or more occurrences of a string.
- • (Concatenation) – joining two patterns
- U (Union) – either/or (can also be written as )

- Precedence Order:



>>Examples:

1. $(0\cup 1)0^*$

- Means: Starts with 0 or 1, followed by any number of 0s
- Language: {w | w starts with 0 or 1, followed by 0s}

2. $(0\cup 1)^*$

- Means: Any combination of 0s and 1s
- Language: {all binary strings}

3. $(0\cup 1)1$

- Means: Either 0 or 1, followed by 1
- Language: {w | w ends with 1, and starts with 0 or 1}

Designing Regular Expressions

 **Example1:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
 “Language Accepting **Strings of Length Exactly Two**”

Step 1: Understand the Alphabet	Step 2: Understand the Language Requirement
<u>We are working with:</u> $\Sigma = \{a, b\}$ This means only the characters a and b can appear in our strings.	>>The language accepts only strings of length exactly two. This means strings like: • aa • ab • ba • bb Only these four strings are valid. Any string with length $\neq 2$ is not allowed.
Step 3: List All Valid Strings	Step 4: Write the Regular Expression
>>As we saw above, valid strings of length 2 over $\{a, b\}$ are: • aa • ab • ba • bb	$\therefore R = (aa \cup ab \cup ba \cup bb)$ Or $\therefore R = (aa \mid ab \mid ba \mid bb)$ Or $\therefore R = (aa + ab + ba + bb)$

 **Example2:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
 “which accepts **all strings of length at least two**.”

Step 1: Understand the Alphabet	Step 2: Understand the Language Requirement
<u>We are working with:</u> $\Sigma = \{a, b\}$ This means only the characters a and b can appear in our strings.	>>The language accepts all strings of length at least two This means: • The string must be two or more characters long. • This includes all strings of length 2, 3, 4, and so on.
Step 3: List All Valid Strings	Step 4: Write the Regular Expression
>> For strings of length at least two: • Valid examples: aa, ab, ba, bb, aaa, aab, abb, baa, bab, bba, bbb, etc. • Invalid examples (strings of length less than two): a, b	$\therefore R = (a \mid b)(a \mid b)(a \mid b)^*$

Explanation:

- $(a \mid b)$: The first character can be either a or b.
- $(a \mid b)$: The second character can also be either a or b.
- $(a \mid b)^*$: After the second character, any number (including zero) of a or b characters can follow. This makes the string longer than two characters if needed.



Example3: Write the Regular Expression of the following languages, where $\Sigma = \{a, b\}$,
“which accepts **Strings of Length At Most Two.**”

Step 1: Understand the Alphabet	Step 2: Understand the Language Requirement
<u>We are working with:</u> $\Sigma = \{a, b\}$ This means only the characters a and b can appear in our strings.	>>The language accepts all strings of length at Most two. This means: • Length 0 (the empty string, denoted by ϵ) • Length 1 (like a, b) • Length 2 (like aa, ab, ba, bb)
Step 3: List All Valid Strings	Step 4: Write the Regular Expression
>> For strings of length at Most two: • ϵ • a • b • aa • ab • ba • bb	$\therefore R = (\epsilon \cup a \cup b \cup aa \cup ab \cup ba \cup bb)$

Practice Problem

 **Problem1:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
“Language Accepting **Strings of Length Exactly Two**”

→ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{aa, ab, ba, bb\}$

$$\therefore R = (aa \cup ab \cup ba \cup bb)$$

Or

$$\therefore R = (aa \mid ab \mid ba \mid bb)$$

Or

$$\therefore R = (aa + ab + ba + bb)$$

 **Problem2:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
“which accepts **all strings of length at least two.**”

→ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{aa, ab, ba, bb, aaa, aab, abab..., bbaa..., \}$

$$\therefore R = (a \cup b) (b \cup a) (a \cup b)^*$$

 **Problem3:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
“which accepts **all strings of length at Most two.**”

→ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{\epsilon, a, b, aa, bb, ab, ba\}$

$$\therefore R = (\epsilon \cup a \cup b \cup aa \cup ab \cup ba \cup bb)$$

 **Problem4:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
“ $L = \{w \mid \text{each } a \text{ in } w \text{ is followed by at least two } b\}$ ”

→ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{abb, b...abb, abbb..., abbabb..., abbbabb...\}$

$$\therefore R = b^* abb b^* (abb b^*)^*$$

Let's check whether the string “**abbabb**” matches the regular expression for the language:

1. **Breakdown of abbabb:**

- The string can be grouped as: a b b a b b
(Each a is followed by exactly two bs, which satisfies the condition.)

2. **Matching with $b^* a b b b^* (a b b b^*)^*$:**

- $b^* \rightarrow$ Matches 0 leading bs (empty).
- $a b b b^* \rightarrow$ Matches a b b (the first b^* matches 0 trailing bs here).
- $(a b b b^*)^* \rightarrow$ Matches the remaining a b b (again, b^* matches 0 trailing bs).
- **Result:** The entire string abbabb is matched.

 **Problem5:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
“ $L = \{w \mid w \text{ contains odd number of } b\}$ ”

→ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{b, a...b, ba..., bababa..., ba...ba...ba...\}$

$$\therefore R = a^* b (a^* b a^* b a^*)^*$$

Let's check whether the string “ ababab ” matches the regular expression for the language:
$\therefore R = a^*b (a^*b a^*b a^*)^*$
<u>Attempt to match “abab”:</u> <u>First b:</u> a^* matches a. - b matches the first b (count = 1). Remaining string: abab <u>First repetition of $(a^*b a^*b a^*)^*$:</u> a^* matches a. - b matches the second b (count = 2). a^* matches a. - b matches the second b (count = 3). The remaining part of the group $(a^*b a^*b a^*)^*$ is optional (due to $*$), so the match succeeds.
Its correct because it accepts abab (and other even-b strings).

 **Problem5:** Write the Regular Expression of the following languages, where $\Sigma = \{a,b\}$,
 “ $L = \{w \mid w \text{ contains } \text{abab} \text{ substring}\}$ ”

➔ **Here,**

» $\Sigma = \{a,b\}$,

» $L(M) = \{\text{abab}, a..abab, b..abab, \text{abbaba}..., \text{abbabb}...\}$

$$\therefore R = (a \cup b)^* \text{abab} (a \cup b)^*$$

Explanation:	
$(a b)^* \rightarrow$ Matches any sequence of as and bs (including none) before "abab". - abab $\rightarrow$ Matches the exact required substring. $(a b)^* \rightarrow$ Matches any sequence of as and bs (including none) after "abab".	
Examples:	
<u>String "abab":</u> $(a b)^*$ matches the empty string (before "abab"). - abab matches the entire string. $(a b)^*$ matches the empty string (after "abab"). ✓ Accepted.	<u>String "abba":</u> - The substring "abab" is missing. ✗ Rejected. <u>String "abbabb":</u> $(a b)^*$ matches the empty string. - abab matches "abab". $(a b)^*$ matches "b". ✓ Accepted.

 **Problem6:** Write the Regular Expression of the following languages, where $\Sigma = \{m, n\}$

- i. “{w | each m in w is followed by at least three n}”
- ii. “{w | w has odd length and starts with m}”
- iii. “{w | w contains ‘mnnmmn’ substring}”

(i)

→ Here,

» $\Sigma = \{m, n\}$,

» $L(M) = \{mnnn, n...mnnn, mnnnn..., mnnmmn... \}$

$$\therefore R = n^* mnnn n^* (mnnn n^*)^*$$

(ii)

→ Here,

» $\Sigma = \{m, n\}$,

» $L(M) = \{m, mmn, mnn, mnnnm, ... \}$

$$\therefore R = m((m \cup n)(m \cup n))^*$$

(iii)

→ Here,

» $\Sigma = \{m, n\}$,

» $L(M) = \{ mnnmmn, mmn...mnnmmn, mnnmmn...mnnnm, ... \}$

$$\therefore R = (m|n)^* mnnmmn (m|n)^*$$

Problem7: Give the **Description** of the following Regular Expressions.

Consider $\Sigma = \{x, y\}$

- i. $((x \cup y)^* xyyxxy)$
- ii. $(x^* y x^* y x^* y y^*)$
- iii. $((x \cup y). ((x \cup y). (x \cup y))^*)$

(i)

→ $(x \cup y)^* xyyxxy$:

- Means: Strings that end with the substring "xyyxxy"..
- Language: $\{w \mid w \text{ is a string over } \Sigma = \{x, y\} \text{ that ends with "xyyxxy"}\}$.

(ii)

→ $x^* y x^* y x^* y y^*$:

- Means: Strings containing minimum three 'y's, where the third 'y' may be followed by any number of additional 'y's..
- Language: $\{w \mid w \text{ is a string over } \Sigma = \{x, y\} \text{ with at least three 'y's, and the third 'y' is followed by zero or more 'y's}\}$.

(iii)

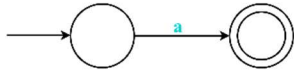
→ $(x \cup y). ((x \cup y). (x \cup y))^*$

- Means: Starts with x or y, and string has odd number of length.
- Language: $\{w \mid w \text{ string starts with x or y, string has odd number of length}\}$

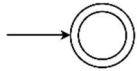
CONVERSION OF RE TO NFA

Formal procedure of Regular Language

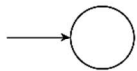
1. $R = a$; $a \in \Sigma$



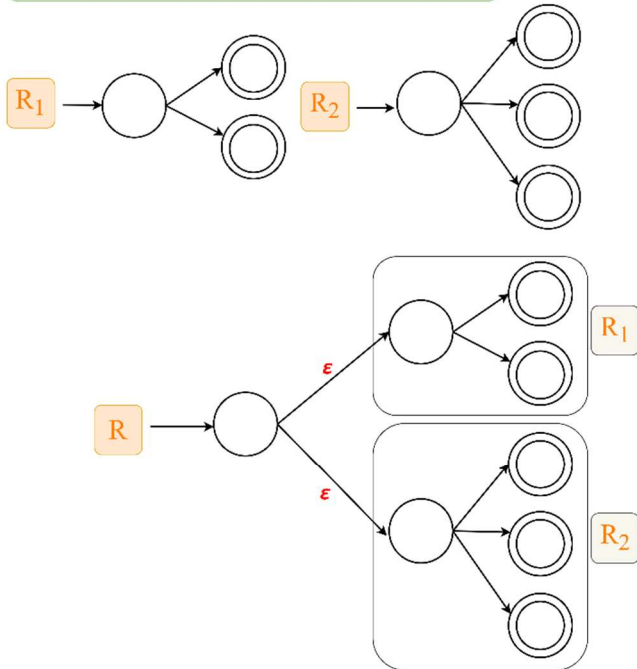
2. $R = \epsilon$; initial state is the final state



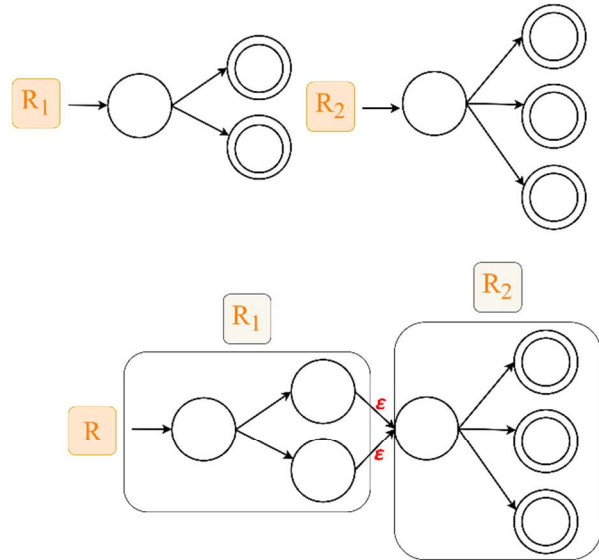
3. $R = \phi$



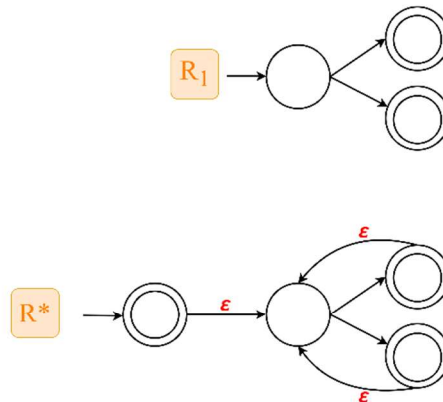
4. $R = R_1 \cup R_2$ or $R = R_1 | R_2$ or $R = R_1$ or R_2



5. $R = R_1 \circ R_2$ or $R = R_1 R_2$ or $R = R_1$ Followed by R_2

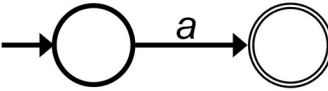
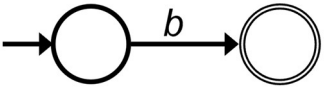
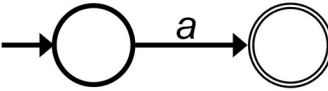
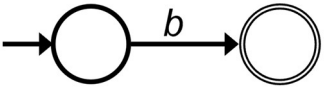
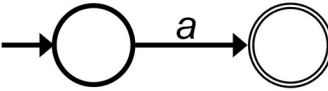
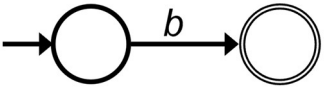
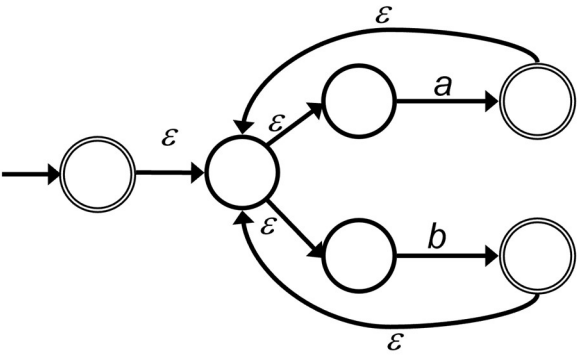
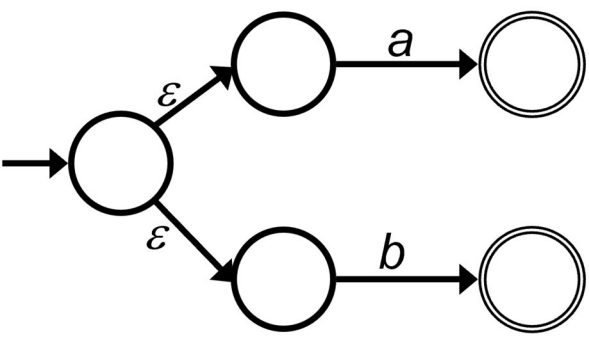
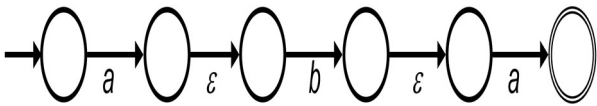
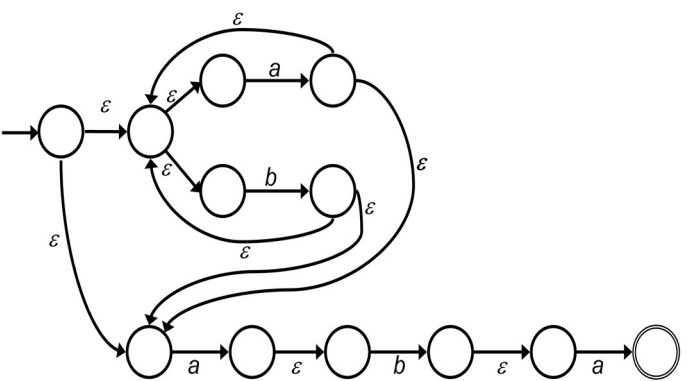


6. $R^* = R_1^*$

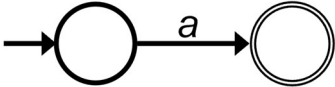
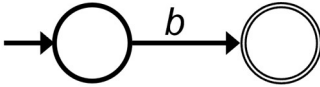
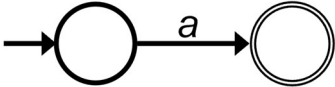
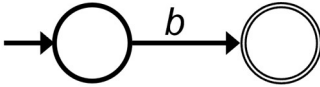
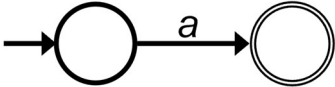
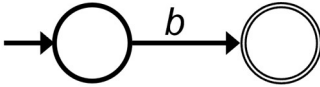
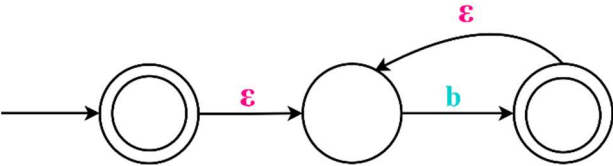
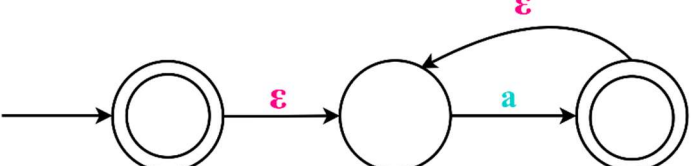
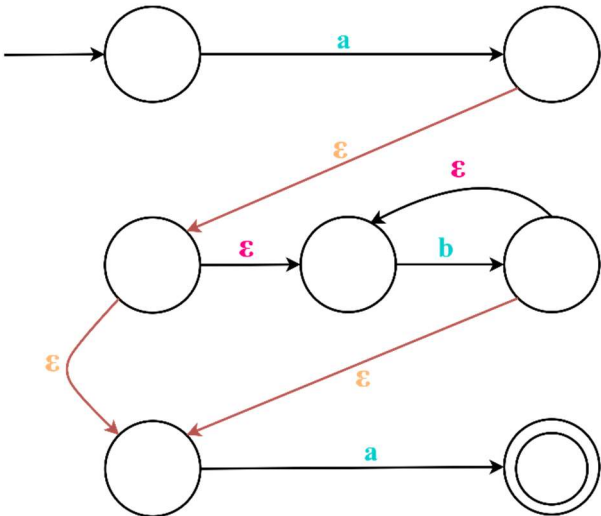
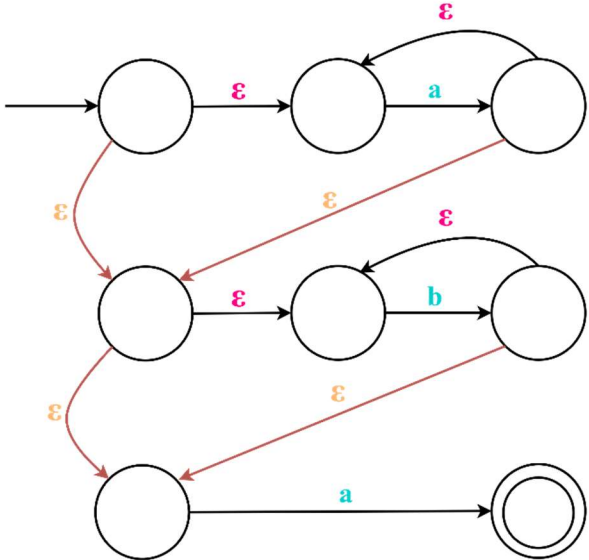


Practice Problem

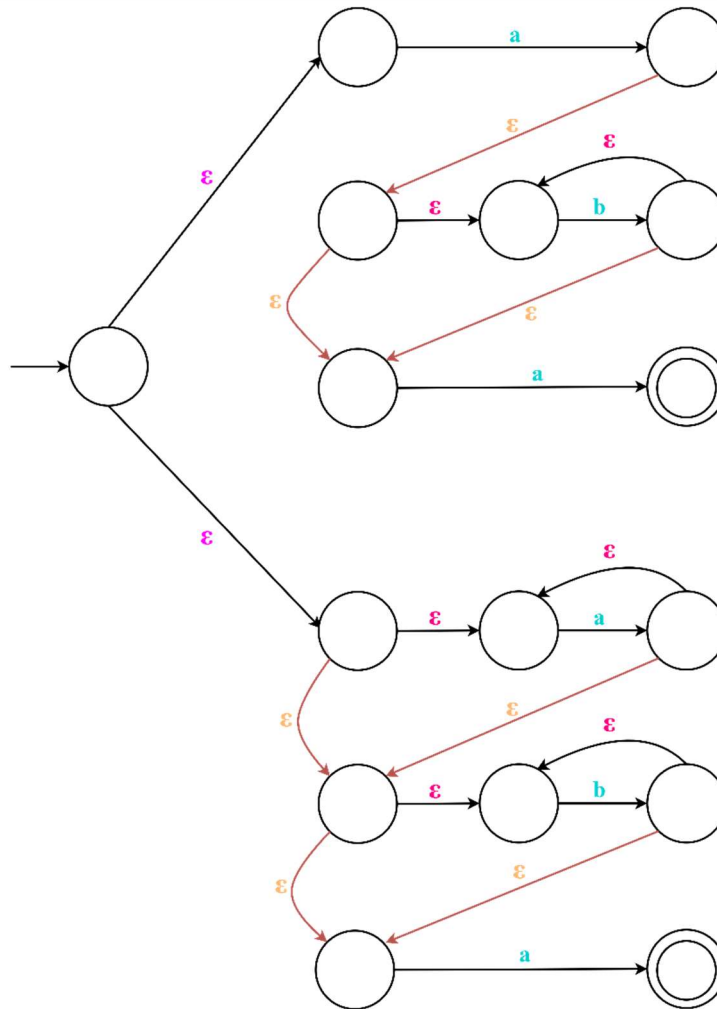
 **Problem1:** Convert the following Regular Expression into equivalent NFA using formal procedure. “ $(a \cup b)^* aba$ ”.

Step 1: Parse the RE into Subexpressions >> The RE “$(a b)^*aba$” can be decomposed as: 1. $(a b)^*$ 2. a 3. b 4. a We'll <u>build NFAs for each component</u> and then concatenate them.	Step 2: Build NFAs for Basic Symbols <table border="1"> <thead> <tr> <th>NFA for a</th><th>NFA for b</th></tr> </thead> <tbody> <tr> <td></td><td></td></tr> </tbody> </table>	NFA for a	NFA for b		
NFA for a	NFA for b				
					
Step 4: NFA for $(a b)^*$ (Star) 	Step 3: NFA for $(a b)$ (Union) 				
Step 5: NFA for “aba” (Concatenation) 	Step 6: Combine $(a b)^*$ and aba (Concatenation) 				

 **Problem2:** Convert the following Regular Expression into equivalent NFA using formal procedure. “**ab*a U a*b*a**”.

Step 1: Parse the RE into Subexpressions >> The RE “ab*a U a*b*a” can be decomposed as: 1. ab*a 2. a*b*a 3. a 5.a* 4. b 6.b* We'll <u>build NFAs for each component</u> and then concatenate them.	Step 2: Build NFAs for Basic Symbols <table border="1"> <thead> <tr> <th>NFA for a</th><th>NFA for b</th></tr> </thead> <tbody> <tr> <td></td><td></td></tr> </tbody> </table>	NFA for a	NFA for b		
NFA for a	NFA for b				
					
Step 4: NFA for b* (Star) 	Step 3: NFA for a* (Star) 				
Step 5: NFA for ab*a (Concatenation) 	Step 6: NFA for a*b*a (Concatenation) 				

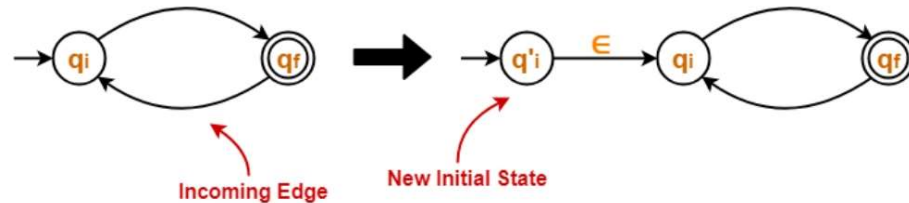
Step 7: Final NFA for $ab^*a \cup a^*b^*a$ (Union)



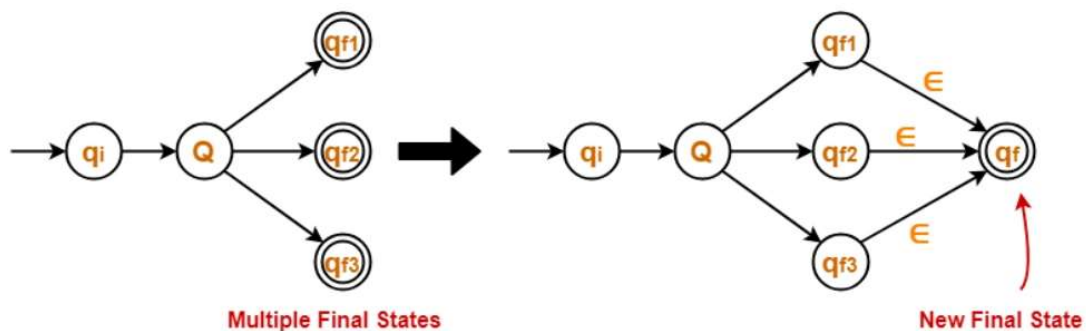
CONVERSION OF DFA TO REG

State Elimination Method & Conversion of GNFA

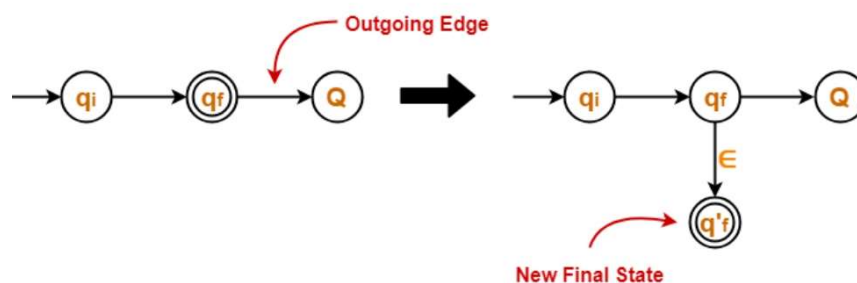
Rule1: There must be only one initial state, and there should not be any incoming edges to it.



Rule2: There must be only one final state.



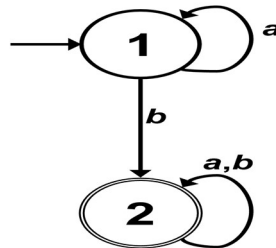
Rule3: From the final state, there should be no outgoing edges.



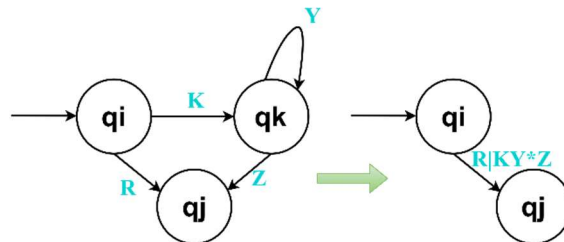
Rule4: After verifying rules 1 to 3, all intermediate states are eliminated one by one. We can eliminate these intermediate states in any order.

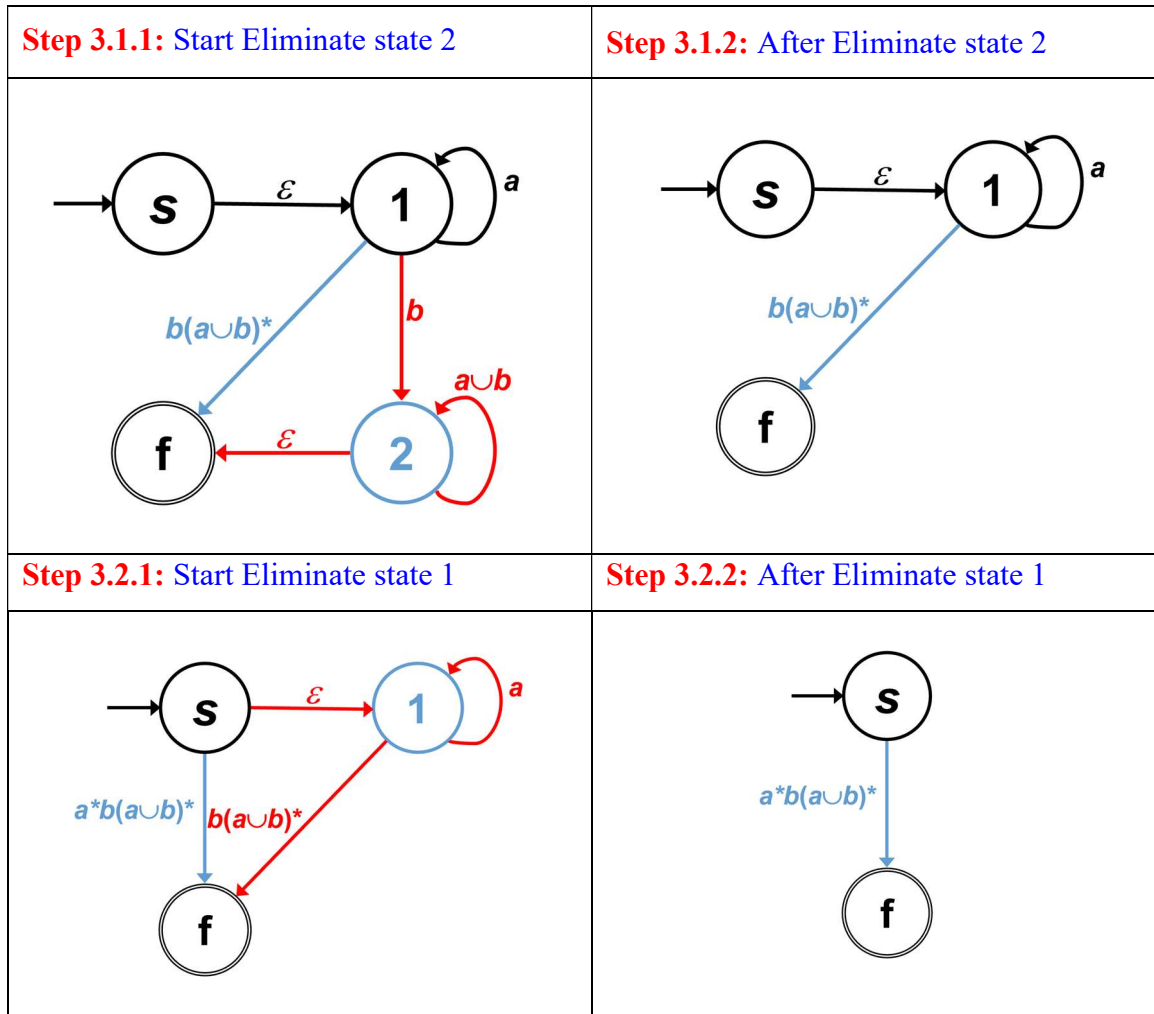
Step-by-Step Conversion of DFA to Regular Expression (RE)

 **Problem1:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.

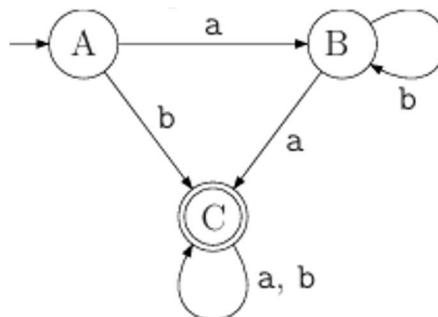


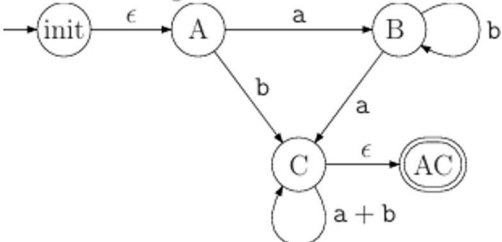
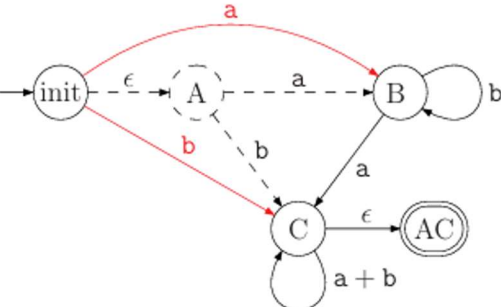
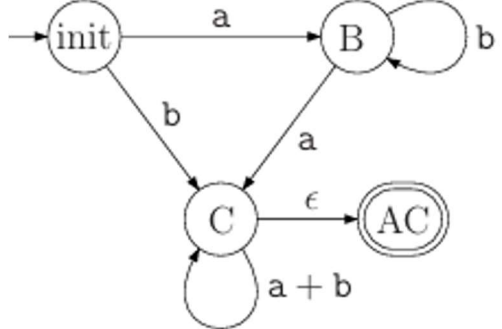
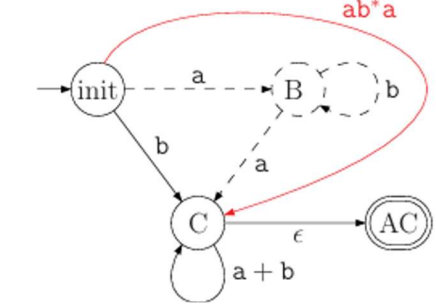
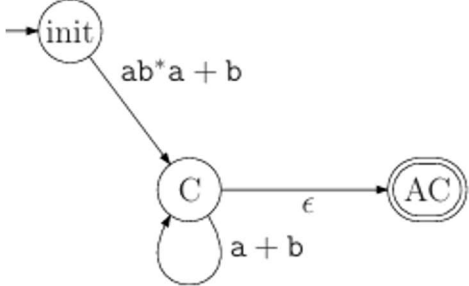
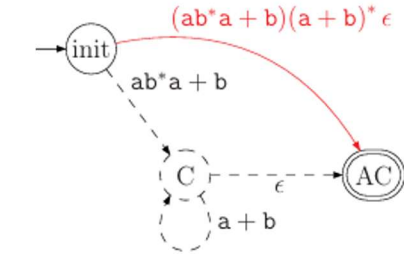
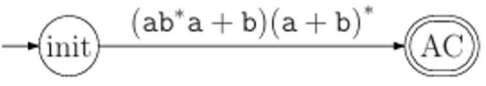
Step1: Represent the DFA	Step2: Converting the DFA to GNFA
➤ $Q = \{1, 2\}$, ➤ $\Sigma = \{a, b\}$, ➤ $q_0 = 1$, ➤ $F = \{2\}$	»Add a new unique start state q_s and final state q_f with ϵ-transitions: ○ $q_s \xrightarrow{\epsilon} q_0$ ○ All original final states $q_f \xrightarrow{\epsilon}$ new q_f »Ensure only one final state (merge if multiple exist).
	<pre> graph LR s((s)) -- ε --> 1((1)) 1 -- a --> 1 1 -- b --> 2((2)) 2 -- "a,b" --> 2 2 -- ε --> f(((f))) </pre>
Step 3: Eliminate States One by One	
»General Rule for Eliminating State q_k: For every pair of states q_i and q_j: - Existing transition: $q_i \xrightarrow{R} q_j$ - Incoming to q_k: $q_i \xrightarrow{X} q_k$ - Loop on q_k: $q_k \xrightarrow{Y} q_k$ - Outgoing from q_k: $q_k \xrightarrow{Z} q_j$ After eliminating q_k add a new transition: $q_i \xrightarrow{R XY^*Z} q_j$	



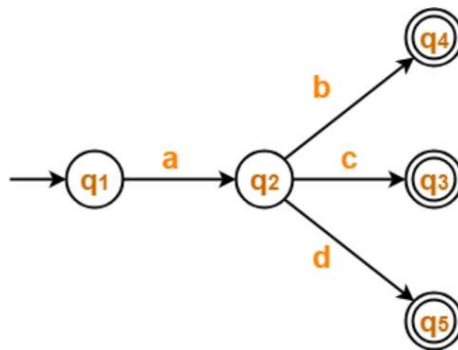


 **Problem2:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



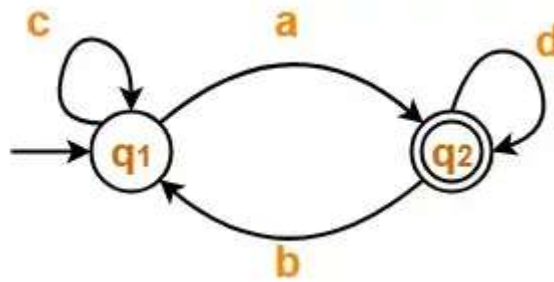
Step1: Represent the DFA ➤ $Q = \{A, B, C\}$, ➤ $\Sigma = \{a, b\}$, ➤ $q_0 = A$, ➤ $F = \{C\}$	Step2: Converting the DFA to GNFA 
Step 3: Eliminate States One by One	
Step 3.1.1: Start Eliminate state A 	Step 3.1.2: After Eliminate state A 
Step 3.2.1: Start Eliminate state B 	Step 3.2.2: After Eliminate state B 
Step 3.3.1: Start Eliminate state C 	Step 3.3.2: After Eliminate state C 

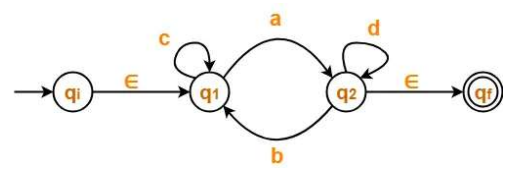
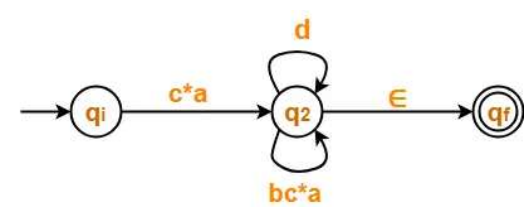
 **Problem3:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



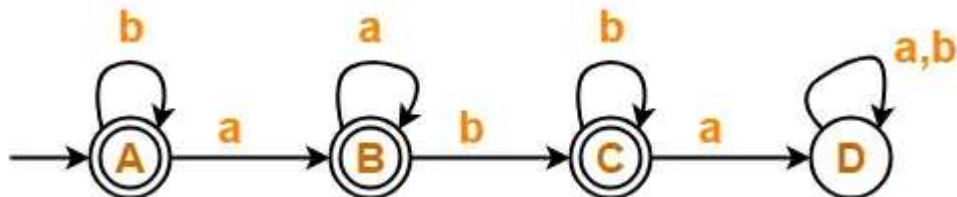
Step1: Represent the DFA	Step2: Converting the DFA to GNFA
➤ $Q = \{q_1, q_2, q_3, q_4, q_5\}$, ➤ $\Sigma = \{a, b, c, d\}$, ➤ $q_0 = q_1$, ➤ $F = \{q_3, q_4, q_5\}$	
Step 3: Eliminate States One by One	
Step 3.1: Eliminate state q4	Step 3.2: Eliminate state q3
Step 3.3: Eliminate state q5	Step 3.4: Eliminate state q2

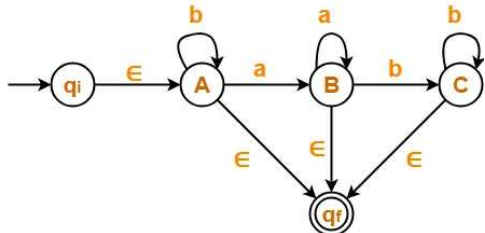
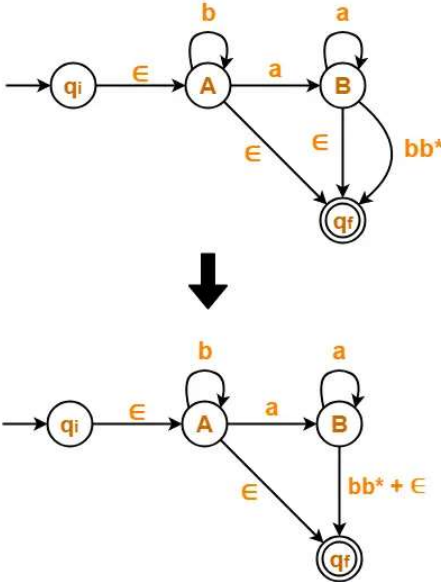
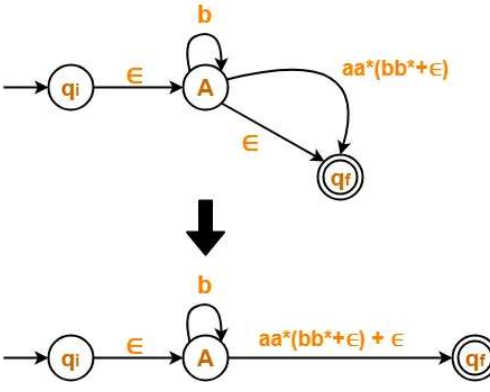
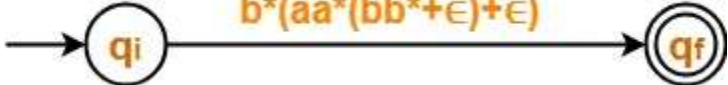
 **Problem4:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



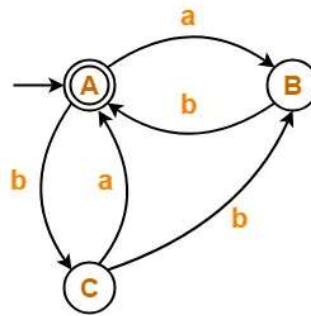
Step1: Represent the DFA ➤ $Q = \{q_1, q_2\}$, ➤ $\Sigma = \{a, b, c, d\}$, ➤ $q_0 = q_1$, ➤ $F = \{q_2\}$	Step2: Converting the DFA to GNFA 
Step 3: Eliminate States One by One	
Step 3.1: Eliminate state q1 	Step 3.2: Eliminate state q2 

 **Problem5:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



Step1: Represent the DFA ➤ $Q = \{A, B, C, D\}$, ➤ $\Sigma = \{a, b\}$, ➤ $q_0 = A$, ➤ $F = \{A, B, C\}$	Step2: Converting the DFA to GNFA 
Step 3: Eliminate States One by One	
Step 3.1: Eliminate state C 	Step 3.2: Eliminate state B 
Step 3.3: Eliminate state A	
	

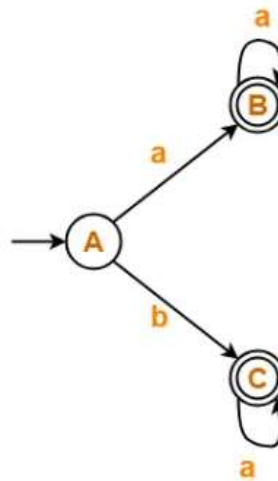
 **Problem6:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



Step1: Represent the DFA	Step2: Converting the DFA to GNFA
➤ $Q = \{A, B, C\}$, ➤ $\Sigma = \{a, b\}$, ➤ $q_0 = A$, ➤ $F = \{A\}$	
Step 3: Eliminate States One by One	
Step 3.1: Eliminate state B	

Step 3.2: Eliminate state C	Step 3.3: Eliminate state A

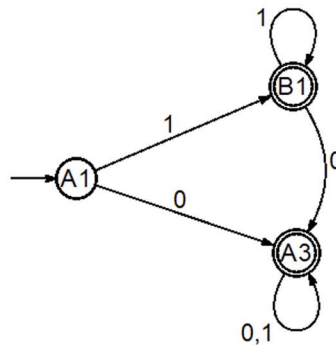
 **Problem7:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



Step1: Represent the DFA	Step2: Converting the DFA to GNFA
➤ $Q = \{A, B, C\}$, ➤ $\Sigma = \{a, b\}$, ➤ $q_0 = A$, ➤ $F = \{B, C\}$	

Step 3: Eliminate States One by One	
Step 3.1: Eliminate state B	Step 3.2: Eliminate state q2

 **Problem8:** Convert the following DFA into equivalent Regular Expression using Formal Procedure.



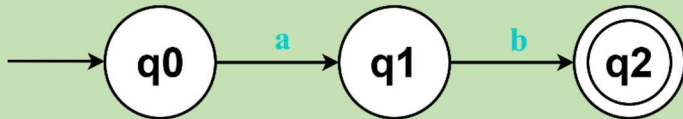
Step1: Represent the DFA	Step2: Converting the DFA to GNFA
➤ $Q = \{A1, A3, B1\}$, ➤ $\Sigma = \{0, 1\}$, ➤ $q_0 = A1$, ➤ $F = \{A3, B1\}$	

Step 3: Eliminate States One by One	
Step 3.1.1: Start Eliminate state A1	Step 3.1.2: After Eliminate state A1
Step 3.2.1: Start Eliminate state B1	Step 3.2.2: After Eliminate state B1
Step 3.3.1: Start Eliminate state A3	Step 3.3.2: After Eliminate state A3

Basics of Conversion: Practice Exercises

Conversion 1:

DFA

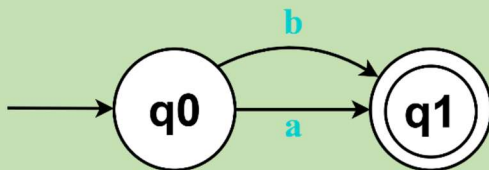


RE

$$\therefore R = ab$$

Conversion 2:

DFA

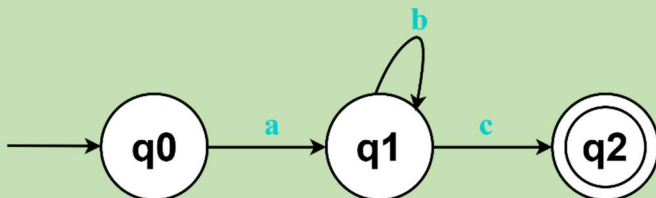


RE

$$\therefore R = a|b$$

Conversion 3:

DFA

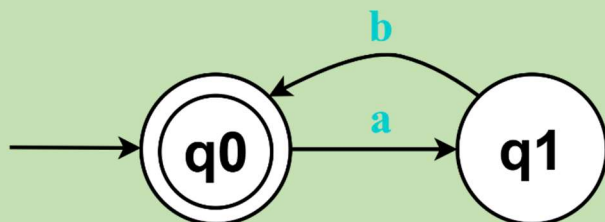


RE

$$\therefore R = ab^*c$$

Conversion 4:

DFA

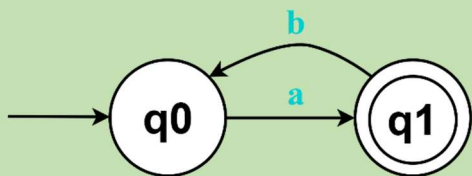


RE

$$\therefore R = (ab)^*$$

Conversion 5:

DFA

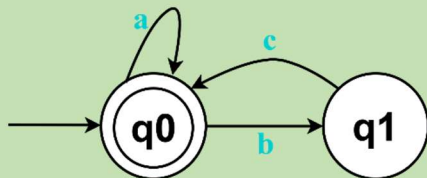


RE

$$\therefore R = a (ba)^*$$

Conversion 6:

DFA

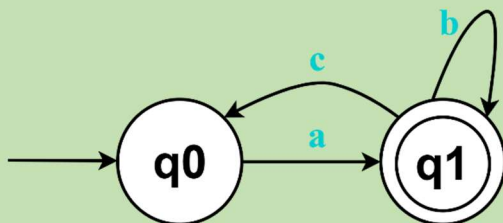


RE

$$\therefore R = (a^* | (bc)^*)^*$$

Conversion 7:

DFA

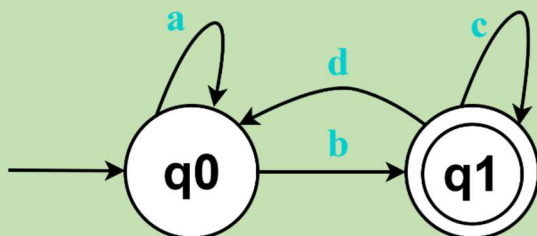


RE

$$\therefore R = a(b^* | (ca)^*)^*$$

Conversion 8:

DFA



RE

$$\therefore R = a^* | b(c^* | (d(a^* | b))^*)^*$$